MIL-M-49241A(CR)
15 June 1987
SUPERSEDING
MIL-M-49241(CR)
17 November 1981

MILITARY SPECIFICATION

MULTIPLEXER, TD-1236/G

This specification is approved for use within the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of the Multiplexer, TD-1236/G. This unit is one of the electronic units that comprise the Digital Group Multiplexer (DGM) family of cable system equipment developed by the Joint Tactical Command, Control and Communications Agency (JTC^3A, formerly TRI-TAC). See 6.1 for intended use and functional description of the unit.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-LC, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation-packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics Measurements of

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MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Gerator Set Family
MIL-STD-2073-2	Packaging Requirement Codes
MIL-STD-45662	Calibration System Requirements

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability for the Family of Digital Group Mulitplexer Equipments
CR-CX-0174-001	TEMPEST for the Family of Digital Group Multiplexer Equipment

Joint Tactical Command, Control and Communications Agency

Interface Control Document ICD-003	Framing and Synchronization Protocol

DRAWINGS

Communications-Electronics Command

DL-SM-B-875929	Multiplexer TD-1236/G
SM-D-884544	Circuit Card Assembly (CCA), Group MUX/DEMUX
SM-D-875942	CCA, Timing
SM-D-875944	CCA, Super Group A
SM-D-875945	CCA, Microcontroller
SM-D-875948	CCA, BITE 1
SM-D-876006	CCA, Super Group Timing
SM-D-884528	CCA, Filter Assembly
SM-D-876269	CCA, Super Group B
SM-D-884567	CCA, Multi VDC
SM-D-884579	CCA, AC Input
SM-D-884593	CCA, 5 VDC

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(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The TD-1236/G shall be constructed in accordance with the requirements of this specification and of DL-SM-B-875929, including the requirements for parts, materials and processes therein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirements 30 and 64.

3.3 Parts, materials, and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawing DL-SM-B-875929.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. (See 4.16) Serial numbers are required on each nomenclatured item including Circuit Card Assemblies (CCAs). The numbers shall be consecutively assigned.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled equipment shall be electrically preconditioned for a

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minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational check of 4.7.7 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Balanced equipment-side signal characteristics.

3.7.1.1 Line-to-line voltage levels. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1, and between -0.1 and -4.0 volts for a digital 0, measured into a 100 ohm dummy load.

3.7.1.2 Line-to-ground voltage. (See 4.7.1.2) The magnitude of the line-to-ground voltage shall not exceed 3 volts.

3.7.1.3 Clock/data phasing. (See 4.7.1.3) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either the upward or downward transition of the data signal shall be no more than 20 ns for group rate signals. Times shall be measured at the zero voltage point of the signals.

3.7.1.4 Offset. (See 4.7.1.4) The magnitude of the offset voltage measured between the center tap of a 100 ohm dummy load and ground shall not be greater than 3 volts.

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3.7.1.5 Signal balance. (See 4.7.1.5) The magnitude of the difference between the magnitudes of the digital 1 level and the digital 0 level shall be less than 10% of the larger of the two.

3.7.1.6 Slew rate. (See 4.7.1.6) When 100 feet of RG-108A/U cable terminated in a 78 ohm dummy load is connected to the driver output, the transition between ±40 mV (either direction) shall take no more than 10 ns for group rate signals.

3.7.1.7 Receiver dynamic range. (See 4.7.1.7) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 mV for a digital 1 or -50 mV for a digital 0 is applied to the receiver input terminals.

3.7.1.8 Input load impedance. (See 4.7.1.8) The input impedance shall be 100 ohms ±10%.

3.7.1.9 Clock asymmetry. (See 4.7.1.9) The positive half of the clock waveform shall be equal to half the period ±15 ns for group rate timing signals.

3.7.1.10 External clock/data phasing. (See 4.7.1.10) The phase relationship between the station clock input or the Super Group related timing input (depending on the timing option selected) and the Super Group data output shall be such that the negative to positive transition of the input shall be led by the output data transitions by 62.3 ±40.6 ns ±3 1/2% of a bit interval.

3.7.2 Control signals. (See 4.7.2)

3.7.2.1 Resync command, waveform parameters. (See 4.7.2.1) The resync command shall be put out by the multiplexer as a balanced pulse signal. This signal shall have an idling level line-to-line between -0.3 and -3.0 volts. When resync is necessary, it shall rise to between +0.3 and +3.0 volts line-to-line. These voltages shall be measured into a 6k ohm load. The rise and fall time 10 to 90% shall be less than 3 microseconds and the pulse width shall be greater than 10 microseconds. The TD-1236/G shall be capable of initiating individual and concurrent cryptographic resynchronizations with interfacing KG-81s (TEDs) on any of its group ports and its Super Group port.

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3.7.2.2 Frame squelch interface. (See 4.7.2.2) The multiplexer shall accept a frame squelch signal. When nothing is connected, the multiplexer is permitted to search for frame. When the applied voltage is between +2 and +5V, the multiplexer is permitted to search for frame. When the voltage applied is between +0 and +0.8V, the multiplexer is inhibited from searching for frame.

3.7.3 Multiplexing. (See 4.7.3) The multiplexer shall be capable of multiplexing groups into a Super Group in the combinations shown in tables I, II and III. A 2 kb/s telemetry channel shall be multiplexed with the groups.

TABLE I. TD-1236/G group rates.

Input Bit Rates (kb/s)				Output Bit Rate (kb/s)	Frame Format Switch
Input #1	Input #2	Input #3	Input #4		
2048 4/	2048	0	0	4096 4/	3 (A1)
2048	1024	1024	0	4096	1 (B)
2048	256-1024	0,256-512	0,256-512	4096	0 (C)
1024	1024	256-1024	0,256-1024	4096	2 (A3)
1024	512	512	256-512	4096	2 (A3)
1024	1024	0	0	2048	3 (A1)
1024	512	512	0	2048	1 (B)
1024	128-512	0,128,256	0,128,256	2048	0 (C)
512	512	128-512	0,128,256,512	2048	2 (A3)
512	256	256	128,256	2048	2 (A3)
512	512	512	0	1536 2/	4 (A2)
512	512	0	0	1024	3 (A1)
512	256	256	0	1024	1 (B)
512	128-256	0,128	0,128	1024	0 (C)
256	256	128-256	0,128-256	1024	2 (A3)
256	128	128	128	1024	2 (A3)
512	0	0	0	512*	7 (S.C)
256	256	0	0	512	3 (A1)
256	128	0,128	0	512	1 (B)
128	128	128	0,128	512	2 (A3)

(Table continued next page.)

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TABLE I. TD-1236/G group rates (Continued).

Input Bit Rates (kb/s)				Output Bit	
Input #1	Input #2	Input #3	Input #4	Rate (kb/s)	Frame Format Switch
256	0	0	0	256*	7 (S.C.)
128	128	0	0	256	3 (A1)
128	0	0	0	128*	7 (S.C.)

*The overhead and traffic bit relationship shall be identical to the input overhead and traffic bit relationship.

NOTES:

1. X-Y means X or Y or any standard intermediate bit rate; e.g., 128-1024 means 128, 256, 512 or 1024.

2. For the 512, 512, 512, 0 kb/s input bit rate combination, both 2048 and 1536 kb/s output rates are applicable, and the required output rate shall be switch selectable.

3. 128 kb/s input group rate will only be utilized with a 16 kb/s channel rate.

4. With 16 kb/s channels, the maximum output bit rate is 2048 kb/s (128 channels), and the maximum input bit rate is 1024 kb/s (64 channels).

TABLE II. TD-1236/G group rates.

Input Bit Rates (kb/s)				Output Bit	
Input #1	Input #2	Input #3	Input #4	Rate (kb/s)	Frame Format Switch
2304	2304	0	0	4608	3 (A1)
2304	1152	1152	0	4608 3/	1 (B)
2304	288-1152	0,288-576	0,288-576	4608	0 (C)
1152	1152	288-1152	0,288-1152	4608	2 (A3)
1152	576	576	288-576	4608	2 (A3)
1152	1152	0	0	2304	3 (A1)
1152	576	576	0	2304	1 (B)

(Table continued next page.)

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TABLE II. TD-1236/G group rates (Continued).

Input Bit Rates (kb/s)				Output Bit Rate (kb/s)	Frame Format Switch
Input #1	Input #2	Input #3	Input #4		
1152	144-576	0,144,288	0,144,288	2304	0 (C)
576	576	144-576	0,144,576	2304	2 (A3)
576	288	288	144-288	2304	2 (A3)
576	576	0	0	1152	3 (A1)
576	288	288	0	1152	1 (B)
576 4/	144,288	0,144	0,144	1152	0 (C)
288	288	144,288	0,144,288	1152	2 (A3)
288	144	144	144	1152	2 (A3)
576	0	0	0	576*	7 (S.C.)
288	288	0	0	576	3 (A1)
288	144	0,144	0	576	1 (B)
144	144	144	0,144	576	2 (A3)
288	0	0	0	288*	7 (S.C.)
144	144	0	0	288	3 (A1)
144	0	0	0	144*	7 (S.C.)

*The output overhead and traffic bit relationship shall be identical to the input overhead and traffic bit relationship.

NOTES:

1. X-Y means X or Y or any standard intermediate bit rate; e.g., 144-1152 means 144, 288, 576 or 1152.

2. 144 kb/s input group rate will only be utilized with a 16 kb/s channel rate.

3. With 16 kb/s channels, the maximum output rate is 2304 kb/s (144 channels), and the maximum input bit rate is 1152 kb/s (72 channels).

4. This line restricted to 16 kb/s channel rate.

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TABLE III. TD-1236/G group rates (TD-1233/TTC operation).

Input Bit Rates (kb/s)				Output Bit Rate (kb/s)	Frame Format Switch
Input #1	Input #2	Input #3	Input #4		
144	---	---	---	288 (9 ch)	6 (D2)
144	144	---	---	288 (9 ch)	6 (D2)
144	144	144	---	576 (18 ch)	5 (D1)
144	144	144	144	576 (18 ch)	5 (D1)

NOTE: Bit rates reduced by a factor of two when loop rate is reduced to 16 kb/s.

3.7.4 Framing. (See 4.7.4)

3.7.4.1 Frame acquisition. (See 4.7.4.1) The frame acquisition time is defined as the period from the start of frame search until the multiplexer at each end of a link is capable of restoring traffic in both directions on the link between them. All multiplexers shall acquire frame synchronization within 50/100 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9, in an error environment of 0.1% bit error rate (BER) random errors. All multiplexers shall also acquire frame synchronization within 100/150 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9, in an error environment of 20% BER random bursts interspersed with 0.1% BER random error intervals. The time between the start of successive burst shall be random between 50 msec and 1 second, and the burst length shall be 5% of the time between the start of that burst and the start of next burst. The above times do not include cryptographic resync delays, or any preset delay intended to assist in flywheeling through fades.

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3.7.4.2 Frame maintenance. (See 4.7.4.2) While bit count integrity is maintained, "out of sync" shall not be declared more often than once per 240 hours (on the average) in an error environment of 20% BER bursts interspersed with 0.1% BER random error intervals. The time between the start of successive bursts shall be random between 50 msec and 1 sec, and the burst length shall be 5% of the time between the start of that burst and the start of next burst.

3.7.4.3 Detection of out of sync. (See 4.7.4.3) When bit count integrity is lost and with random data on all channels, a multiplexer shall detect out of sync within 450 to 700 ms at frame rates of 4/2 kb/s, with a minimum probability of 0.9. When out of sync occurs on any link, the multiplexers shall continue to transmit frame pattern on any adjacent connecting links, as appropriate, to maintain frame synchronization of these links.

3.7.4.4 Delayed frame search and false frame lock. (See 4.7.4.4)

a. Delayed frame search. A fixed delay of 500 $\pm$ 50 milliseconds shall be provided.

b. False frame lock. When the Super Group signal is looped-back, the TD-1236/G shall be capable of clearing any false frame lock condition in multiplexers connected to the group ports.

3.7.4.5 Idle filler pattern. (See 4.7.4.5) Channels which are idle because they are unassigned or are not carrying traffic, or traffic channels derived from a link which has lost frame synchronization, shall be set to all ones.

3.7.4.6 Group input buffering. (See 4.7.4.6) Although the TD-1236/G is usually operated in a synchronous system, each one of its four group inputs shall contain a First-In First-Out (FIFO) buffer. The TD-1236/G shall be capable of maintaining bit count integrity for a cumulative shift of $\pm$25 bits between each group input and local reference timing. The FIFO shall reset to center under any one of the following conditions:

a. FIFOs for all four input groups shall re-center when power is applied to the multiplexer.

b. Each group input FIFO for group input ports 2, 3, and 4 shall individually re-center, independently of the other FIFOs, if it overflows or underflows when passing encrypted group

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traffic, i.e., group does not have a 32 or 16 kb/s overhead channel containing a frame pattern as described in ICD-003.

c. Each group input FIFO in any of the four group inputs shall individually re-center, independently of the other FIFOs, as part of the frame acquisition process for that group port when passing plain text group traffic, i.e., group has a 32 or 16 kb/s overhead channel containing a frame pattern as described in ICD-003.

3.7.5 Prime power. (See 4.7.5) The ac primary power shall be 115v ±10% at frequencies of 50, 60, and 400 Hz ±5%, single phase. The power input to the TD-1236/G shall not exceed 75 watts. The ac primary power shall conform to MIL-STD-1332 for Type 1, Class 1 and 2, Mode I, II, and III generation sets.

3.7.6 Built-in test equipment (BITE) and alarms. (See 4.7.6) The built-in test equipment (BITE) shall monitor signals in and out of the unit and internal circuits. The go/no-go fault-location circuits shall isolate failures to a particular printed circuit card or lowest replaceable unit (LRU). Indicating devices shall be provided to insure proper operation of the equipment and to monitor the dc voltages, go/no-go fault isolation, and other critical internal and external signals. The BITE shall operate without interrupting traffic or interfering with normal system operation. Removal of the BITE card shall not affect operation of the unit. Failure alarms shall be provided for channel failure, common power supply failure, and interunit signal malfunctions. Failures detected by the BITE circuitry cause relay contact closures accessible externally at a connector on the rear panel. The S and SA alarm outputs as described below are available.

a. S Alarm. The S alarm shall indicate any failure detected including loss of input power. This alarm shall also provide a visible indication on the front panel except for loss of input power.

b. SA Alarm. The SA alarm shall indicate that the malfunction is within the unit. If the power switch is set to OFF, there shall be no SA Alarm.

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3.7.7 Operational check. (See 4.7.7)

3.7.7.1 Operational data error check. (See 4.7.7.1) There shall be no errors for a period of 10 seconds.

3.7.7.2 Operational alarm monitor check. (See 4.7.7.2) There shall be no alarms generated.

3.7.7.3 Operational telemetry error check. (See 4.7.7.3) There shall be no errors for a period of 30 seconds.

3.7.8 Timing modes. (See 4.7.8) The unit is intended for operation as a synchronous system and shall utilize timing from the sources described in 3.7.8.1 and 3.7.8.2.

3.7.8.1 Station clock timing. (See 4.7.8.1) In this mode, the transmit Super Group shall be timed by an external station clock supplied at the Super Group rate. The group signals shall be synchronous to this clock.

3.7.8.2 Loopback timing. (See 4.7.8.2) In this mode, the transmit Super Group shall be timed by the received Super Group clock. The group signals shall be synchronous to this clock.

3.7.9 Environmental operational test.

3.7.9.1 Environmental group data error rate. (See 4.7.9.1) There shall be no errors for a measurement period of 10 seconds.

3.7.9.2 Environmental telemetry error check. (See 4.7.9.2) There shall be no errors for a measurement period of 30 seconds.

3.7.9.3 Environmental alarm monitor operation. (See 4.7.9.3) No alarms shall be generated, that is:

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a. There shall be open circuits on the S and SA alarm outputs (a short indicates alarm).

b. When the Super Group connector on the auxiliary multiplexer is disconnected, an S and loss of input (LOI) alarm shall be generated (a short indicates an alarm).

3.7.9.4 Sync/frame check. (See 4.7.9.4)

a. When the TD-1236/G is out of frame, it shall generate a resync command pulse (10 us wide) approximately every second.

b. When in frame, no resync command pulse shall be generated.

3.8 Service conditions. (See 4.8) The operational test of 3.7.9 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1) The operational test of 3.7.9 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +63°C (+145°F), and shall withstand exposure (non-operating) to ambient air temperatures as high as +71°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -31.7°C (-25°F), and shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

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3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, the effects of dust particles that may accumulate within the shelter assembly or vehicle.

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Vibration and shock. (See 4.8.7). The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage, over all types of roads and cross-country terrain, and shocks encountered during servicing.

3.9 Electromagnetic interference (EMI). (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4.

CE02	CS02	RE02	RS02
CE03	CS06	RE02.1	RS03
CE04			
CE05			

NOTES:

a. RE02.1 is applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 is applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to dc, 60 Hz and 400 Hz power lines.

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3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The unit shall have a specified Mean-Time-Between-Failure (MTBF), θ_o, of 6000 hours when in the normal modes of operation.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements specified on their respective test requirement specification (TRS) drawings, as referenced in the following subassembly drawings:

CCA	Drawings
Microcontroller*	SM-D-875945
Group Mux/Demux*	SM-D-884544
Timing *	SM-D-875942
Super Group A*	SM-D-875944
Super Group B*	SM-D-875269
Super Group Timing*	SM-D-876006
BITE 1*	SM-D-875948
Filter Assembly	SM-D-884528
Multi Vdc	SM-D-884567
5 Vdc	SM-D-884593
AC Input	SM-D-884579

* This drawing contains automatic test equipment (ATE) acceptance procedures and programs.

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3.13 Operational test. (See 4.7.7) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.7 after all other inspections and tests are completed to determine that the unit is operating prior to packaging. The operational test of group A satisfies this requirement for those units for which this is the last inspection or test.

3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8, shall be applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with the requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

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3.16 Nuclear survivability. (See 4.17) The equipment shall satisfy the nuclear survivability requirements stated in Section 3 of CR-CX-0173-001 with the exception of paragraphs 3.3.1.1 (BLAST) and 3.3.1.2 (THERMAL).

3.17 TEMPEST. (See 4.18) The equipment shall satisfy the TEMPEST requirements of CR-CX-0174-001.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections of 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

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b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection before packaging. (See 4.5)

(2) Packaging inspection. (See 4.14)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 20 first article units of the complete Multiplexer TD-1236/G. The first article quantity shall be as specified in the contract or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table IV and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in-process group A inspection prior to burn-in.

4.3.3 First article data. The first article test plan and test report(s) be as required in the contract or purchase order.

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TABLE IV. First article inspections.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
1. Inspections by Subsidiary	N/A	4.4	Inspections to be performed on all units.												
2. Burn-In	3.6	4.6	Inspections to be performed on all units.												
3. Group A Inspection	(See Table V)		Inspections to be performed on all units.												
4. Group B Inspection	(See Table VI)		Inspections to be performed on all units.												
5. Group C Inspection															
Subgroup 1.															
High Temperature	3.8.1.1	4.8.1.1	1	*											
Low Temperature	3.8.1.2	4.8.1.2	2	*											
Subgroup 2															
EMI	3.9	4.9			1										
TEMPEST	3.17	4.18			2										
Vibration/Shock	3.8.7	4.8.7				1	*								
Subgroup 3															
Humidity (Note c)	3.8.2	4.8.2						1			*				
Subgroup 4															
Altitude	3.8.3	4.8.3							1		*				
Dust(Note c)	3.8.4	4.8.4							2		*				
Salt Fog (Notec)	3.8.5	4.8.5							3		*				
Fungus (Note c)	3.8.6	4.8.6								1					

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TABLE IV. First article inspections (Continued).

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8	9	10	11	12-19	20+
Subgroup 5															
Line-to-Ground Voltage	3.7.1.2	4.7.1.2**										1			
Clock/Data Phasing	3.7.1.3	4.7.1.3										2			
Offset	3.7.1.4	4.7.1.4**										3			
Signal Balance	3.7.1.5	4.7.1.5										4			
Slew Rate	3.7.1.6	4.7.1.6										5			
Input Impedance	3.7.1.8	4.7.1.8										6			
Clock Asymmetry	3.7.1.9	4.7.1.9										7			
External Clock/Data Phasing	3.7.1.10	4.7.1.10										8			
Resync Command Waveform	3.7.2.1	4.7.2.1.1										9			
Multiplexing	3.7.3	4.7.3										10			
Frame Acquisition Time	3.7.4.1	4.7.4.1										11	*		
Frame Maintenance	3.7.4.2	4.7.4.2**										12			
Detection of Out of Sync	3.7.4.3	4.7.4.3										13			
Delayed Frame Search	3.7.4.4a.	4.7.4.4										14			
False Frame Lock	3.7.4.4b.	4.7.4.4.1**										15			
Idle Filler Pattern	3.7.4.5	4.7.4.5**										16			
Group Input Buffering	3.7.4.6	4.7.4.6**										17			
Prime Power	3.7.5	4.7.5										18			
Timing Mode Tests	3.7.8	4.7.8										19			
Dimensional Interchangeability	3.11	4.11										20			
6. Group D inspection															
Reliability	3.10	4.10												8 Units	

NOTES:

a. The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

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b. The number in the unit columns in the table covered by inspections 5 and 6 specify the order of inspections for the indicated unit. An * indicates the use of an auxiliary unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. The quantity of units shall be as specified in the contract or purchase order. All first article units shall be subjected to group A and group B inspections.

e. A ** indicates one time engineering test to be covered by a separate engineering report not part of first article test report.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this inspection, and the inspection requirement specified in the contract or purchase order.

	Where required
Printed Wiring	3.2.2
Semiconductor Devices and Integrated Circuits	3.2.3
Parts, Materials, and Processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table V shall be inspected for conformance to the inspections specified in table V.

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4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table V. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table V shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table V.

b. Only for spares that are not installed into complete equipment, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table V (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspections of step a above shall be assembled into the end equipments, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and an end equipment 100% visual/mechanical and electrical inspections in accordance with table V in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table V. Group A electrical inspection shall be performed in the order shown in table V.

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TABLE V. Group A inspection.

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
VISUAL-MECHANICAL (Including Spare Assemblies)				
Microcontroller	3.15	4.13	2.5%	4.0%
Group MUX/DEMUX	3.15	4.13	2.5%	4.0%
MULTI VDC	3.15	4.13	2.5%	4.0%
5 VDC	3.15	4.13	2.5%	4.0%
AC Input	3.15	4.13	2.5%	4.0%
Timing	3.15	4.13	2.5%	4.0%
Super Group A	3.15	4.13	2.5%	4.0%
Super Group B	3.15	4.13	2.5%	4.0%
Super Group Timing	3.15	4.13	2.5%	4.0%
BITE 1	3.15	4.13	2.5%	4.0%
Filter Assembly	3.15	4.13	2.5%	4.0%
Case & Cover	3.15	4.13	2.5%	10.0%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
Microcontroller	3.12	4.12	1.0%	*
Group MUX/DEMUX	3.12	4.12	1.0%	*
MULTI VDC	3.12	4.12	1.0%	*
5 VDC	3.12	4.12	1.0%	*
AC Input	3.12	4.12	1.0%	*
Timing	3.12	4.12	1.0%	*
Super Group A	3.12	4.12	1.0%	*
Super Group B	3.12	4.12	1.0%	*
Super Group Timing	3.12	4.12	1.0%	*
BITE 1	3.12	4.12	1.0%	*
Filter Assembly	3.12	4.12	1.0%	*
ELECTRICAL				
BITE and Alarms	3.7.6	4.7.6	1.0%	*
Operational Tests	3.7.7	4.7.7	1.0%	*

*All electrical defects are major.

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4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table VI and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B AQL shall be as shown in table VI and the inspection level shall be S-4.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in the order listed in table VI.

TABLE VI. Group B inspections.

Inspection	Rqmt Para	Insp Para	AQL
Line-to-Line Voltage Levels	3.7.1.1	4.7.1.1	4.0%
Receiver Dynamic Range	3.7.1.7	4.7.1.7	4.0%
Resync Voltage	3.7.2.1	4.7.2.1	4.0%
Frame Squelch Interface	3.7.2.2	4.7.2.2	4.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units from lots that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table VII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature in that order.

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4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to vibration and shock testing in that order. The unit selected every eighth month shall be subjected to EMI and TEMPEST testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to altitude, dust and salt fog testing, and one unit shall be subjected to fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table VII.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order.

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TABLE VII. Group C inspection.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High temperature	3.8.1.1	4.8.1.1
Low temperature	3.8.1.2	4.8.1.2
Subgroup 2		
Vibration	3.8.7.1	4.8.7.1
Shock	3.8.7.2	4.8.7.2
Basic design		4.8.7.2.1
Bench handling		4.8.7.2.2
EMI	3.9	4.9
TEMPEST	3.17	4.18
Subgroup 3		
Humidity	3.8.2	4.8.2 (Note 1)
Subgroup 4		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)
Subgroup 5		
Clock/Data Phasing	3.7.1.3	4.7.1.3
Signal Balance	3.7.1.5	4.7.1.5
Slew Rate	3.7.1.6	4.7.1.6
Input Impedance	3.7.1.8	4.7.1.8
Clock Asymmetry	3.7.1.9	4.7.1.9
External Clock/Data Phasing	3.7.1.10	4.7.1.10
Resync Command Waveform	3.7.2.1	4.7.2.1.1
Multiplexing	3.7.3	4.7.3
Frame Acquisition Time	3.7.4.1	4.7.4.1
Detection of Out of Sync	3.7.4.3	4.7.4.3
Delayed Frame Search	3.7.4.4.a	4.7.4.4
Prime Power	3.7.5	4.7.5
Timing Mode Tests	3.7.8	4.7.8
Dimensional Inter-changeability	3.11	4.11

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.5.4 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table VIII and shall be performed on units from lots which have been subjected to and have met group A and group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE VIII. Group D inspection.

Inspection	Reqt Para	Insp Para
Reliability	3.10	4.10

4.5.5.1 Sampling for group D inspection. Eight units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsequent group D inspections, eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

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4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected as shown in figure 1. Test conditions of 4.10.3 shall apply except that the duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred. The requirements of 3.6 shall be met.

a. Set UUT 1 as follows:

Group Rate:	9(1152 kb/s)
Frame:	Yes
Timing:	Looped
Super Group Rate:	14(4608 kb/s)
Channel Rate:	32 kb/s
Format:	2

b. Set remaining UUTs as follows:

Group Rate:	9(1152 kb/s)
Frame:	Yes
Timing:	Station Clock
Super Group Rate:	14(4608 kb/s)
Channel Rate:	32 kb/s
Format:	2

c. The S, SA alarms shall be latched to detect any alarm occuring during the 48 hours burn-in period.

d. After burn-in, perform the operational test of 4.7.7 on each unit.

4.7 Electrical. (See 3.7) Unless otherwise specified all tests shall be performed at 115V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810D unless otherwise specified. The electrical test equipment listed in table IX or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662A. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table X) shall be provided by the contractor subjected to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) required for test listed in table XI. Table XII identifies the external connections for the TD-1236. Unless otherwise noted, all resistors in the figures are 0.5 watt.

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TABLE IX. Standard test equipment.

Description	Mfg./Model No.(or equivalent)
Electronic Counter (2 required)	H.P. 5248L
Electronic Counter	H.P. 5360A
Volt-Ohm Meter (2 required)	Simpson 260
Wattmeter	Weston 432
Freq synthesizer	H.P. 5105A
Attenuator	H.P. 4437A
AC Voltage	H.P. 403
Attenuator	Kay 3-30
Oscilloscope	Tektronix 7904
Audio Oscillator	H.P. 200 CD
Pulse Generator	H.P. 3310A
DC Power Supply	Sorensen QRD 30-1
AC Power Supply	ELGAR 401-V

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TABLE X. Special test equipment.

Equipment Name	Drawing No	Functional Description
Group Simulator	A3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It generates a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination.
Timing Source	A3051666	The Timing Source provides four independent group outputs, switch-selectable to positions between 72 and 4608 kb/s. Also provided are four independent Station Clock/ Telemetry Clock outputs, switch-selectable to 2, 16, or 32 kHz.
Timing Source	(Alternate)	An alternate Timing Source is shown in figure 2 that may be used in lieu of the above.
DSVT Simulator	A3051668	The DSVT Simulator provides a pseudo-random pattern generator and a correlator with a conditioned diphase interface at 16 and 32 kb/s. An error output is provided for counting errors externally.
Error Inserter	A3051665	The Error Inserter provides a means of injecting a 0.1% continuous BER with a 20% random bursts.

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TABLE XI. Government furnished equipment (GFE).

Equipment	Quantity	Description
Modem, Digital MD-1026()(P)/G	4 ea	The Group Modem (GM) provides electrical signal compatibility between balanced NRZ digital signals on its equipment side and conditioned diphase signals on it line side. This equipment must be configured with conditioned diphase CCAs in all four group slots.
Multiplexer, Remote Loop Group TD-1233()(P)/TTC	1 ea	The Remote Loop Group Multiplexer time division multiplexes/demultiplexes up to 4 analog or conditioned diphase loops into/from a 4 1/2 channel group. The group input/output is a conditioned diphase signal. This equipment must be configured with Loop Modem CCAs in all channel slots.
Generator, Digital Data SG-1139		Alternate for DSVT Simulator

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TABLE XII TD-1236/G interface connections.

Connector	Function	PIN HI	PIN LO	PIN Gnd	Signal Type
Groups 1-4	Group Data In	B	A	M	NRZ
	Group Timing In	D	C	N	SQ Wave
	Group Data Out	F	E	P	NRZ
	Group Timing Out	H	G	R	SQ Wave
Super Group	Super Group Data In	B	A	M	NRZ
	Super Group Timing In	D	C	N	SQ Wave
	Super Group Data Out	F	E	P	NRZ
	Super Group Timing Out	H	G	R	SQ Wave
	Station Clock In	K	J	S	SQ Wave
TED & Alarm	Resync Command Super Group	S	R		Bal. 10 usec pulse width
	Group 1	L	A		Bal. 10 usec pulse width
	Group 2	M	B		Bal. 10 usec pulse width
	Group 3	P	D		Bal. 10 usec pulse width
	Group 4	N	C		Bal. 10 usec pulse width
	Frame Squelch Signal	F			Unbalanced
	Return	E			Unbalanced
	S Alarm	H	G	T	Relay, Alarm= Close contact
	SA Alarm	K	J	U	Relay, Alarm= Close contact
Power		A	C	B	

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4.7.1 Balanced equipment-side signal characteristics. (See 3.7.1)

4.7.1.1 Line-to-line voltage levels. (See 3.7.1.1)

a. Connect the equipment as shown in figure 3.

b. Set the Group Simulator as follows:

Group Rate: 9 (1152 kb/s)	Type: 1
Timing Mode: External	Channel Rate: 32 kb/s
Transmit Format: MSF	Data Select: Random

Set the UUT as follows:

Group Rate: 9 (1152 kb/s)	Super Group Rate: 14 (4608 kb/s)
Frame: Yes	Channel Rate: 32 kb/s
Timing: Station Clock	Format: 2

c. For each of the signals listed in 4.7.1.3c, measure the positive and negative voltage levels line-to-line.

NOTE: To speed up Group Framing, momentarily set the Group Simulator to Frame Acknowledge Request (FAR) and then back to Multiplex Signal Format (MSF).

d. Verify that the requirements of 3.7.1.1 are met for each signal measured.

4.7.1.2 Line-to-line ground voltage. (See 3.7.1.2) Measure the line-to-ground voltage for each line using test configuration in figure 3. Verify that the requirements of 3.7.1.2 are met.

4.7.1.3 Clock/Data phasing. (See 3.7.1.3)

a. Configure the UUT as shown in figure 3 with switch settings as shown in 4.7.1.1.

b. Configure the oscilloscope as shown in figure 4.

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c. For each of the signals given below, measure the delay between the zero voltage points on both the Data Rise and Fall Time Slopes to the zero voltage points of the Timing Rise Slopes as indicated in figure 5.

(1) Group 1 DATA OUT
(2) Group 1 TIMING OUT
(3) Group 2 DATA OUT
(4) Group 2 TIMING OUT
(5) Group 3 DATA OUT
(6) Group 3 TIMING OUT
(7) Group 4 DATA OUT
(8) Group 4 TIMING OUT
(9) SG DATA OUT
(10) SG TIMING OUT

d. Verify that the requirements of 3.7.1.3 are met.

4.7.1.4 Offset. (See 3.7.1.4)

a. Connect the equipment as shown in figure 3 and set the UUT switches as shown in 4.7.1.1.

b. Set the oscilloscope as shown in figure 6.

c. For each of the signals given in 4.7.1.3c, measure the DC Offset Voltage.

d. Verify that the requirements of 3.7.1.4 are met.

4.7.1.5 Signal balance. (See 3.7.1.5)

a. Connect the equipment as shown in figure 3.

b. Set the UUT switches as shown in 4.7.1.1.

c. For each of the signals given in 4.7.1.3.c, measure the positive and negative voltage levels line-to-line.

d. Compare the positive and negative values. Determine the larger of the two values.

e. Compare the magnitude of the difference between the positive and negative values to 10% of the larger value.

f. Verify that the magnitude of the differences are within the specified requirements of 3.7.1.5.

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4.7.1.6 Slew rate. (See 3.7.1.6)

a. Connect the equipment as shown in figure 3.

NOTE: All measurements made during this test shall be through 100ft of RG/108 cable. The value for R shall be 39 ohms ±1% instead of 50 ohms ±1%.

b. Set the UUT switches as shown in 4.7.1.1.

c. For each of the signals given in 4.7.1.3c, measure the time between the +40mV and the -40mV points around the two voltage level of the rise and fall slopes as indicated in figure 7.

d. Verify that the requirements of 3.7.1.6 are met.

4.7.1.7 Receiver dynamic range. (See 3.7.1.7)

a. Connect the equipment as shown in figure 8. A variable attenuator will be connected in the normal Group, Super Group and Station Clock signal paths during the course of the test.

b. Set the UUT switches as in 4.7.1.1. For each of the signals given below, connect the attenuator in the signal path and adjust the attenuator using the oscilloscope to give a ±50 mV peak signal from the Group Simulator.

c. Observe the Group Simulator for a 10-second period. Verify that no errors are generated.

Group 1 DATA/TIMING
Group 2 DATA/TIMING
Group 3 DATA/TIMING
Group 4 DATA/TIMING
Super Group DATA/TIMING
Station Clock

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4.7.1.8 Input load impedance. (See 3.7.1.8)

a. Connect the equipment as shown in figure 9.

b. Connect the test setup in sequence to the signal inputs given below.

c. For each signal input given below, adjust the audio oscillator at a frequency of 1.0 kHz for an Output Voltage of 1.0 volt as measured by the V_1 Voltmeter; then measure and record the V_2 voltage.

d. Calculate the Input Impedance of each input using the formula:

$$R = 100 \frac{V_2}{V_1}$$

e. Verify that the calculated impedance meets the requirements of 3.7.1.8.

(1) Grp 1 Data In
(2) Grp 1 Tim In
(3) Grp 2 Data In
(4) Grp 2 Tim In
(5) Grp 3 Data In
(6) Grp 3 Tim In
(7) Grp 4 Data In
(8) Grp 4 Tim In
(9) SG Data In
(10) SG Tim In
(11) Sta Clk In

4.7.1.9 Clock asymmetry. (See 3.7.1.9)

a. Connect the equipment as shown in figure 3 with switch settings as shown in 4.7.1.1.

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b. For each of the signals given below, measure the time (A) between the rise and fall slopes at the ZERO VOLTAGE POINTS; then, measure the time (B) between two successive rise slopes at the ZERO VOLTAGE POINTS as indicated in figure 10.

(1) Group 1 Timing
(2) Group 2 Timing
(3) Group 3 Timing
(4) Group 4 Timing
(5) SG Timing

c. Verify that the specified requirements of 3.7.1.9 are met.

4.7.1.10 External clock/data phasing. (See 3.7.1.10)

a. Connect the equipment as shown in figure 3. Set the UUT switches as in 4.7.1.1.

b. Connect the oscilloscope as shown in figure 11.

c. Connect the Timing Source to Station Clock Input.

d. Measure the delay between the zero crossing of the negative-to-positive transition of the Timing Source signal and the zero crossing of any data transition as in figure 5.

e. Connect the Timing Source to the Receive Timing input and set the UUT TIMING switch to looped.

f. Repeat steps d and e.

g. Verify that the requirements of 3.7.1.10 are met for both conditions.

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4.7.2 Control signals. (See 3.7.2)

4.7.2.1 Resync command voltage. (See 3.7.2.1)

a. Connect the equipment as shown in figure 13 with the switch settings as shown in 4.7.1.1 and the Group Simulator providing a No Frame transmit format and looped timing.

b. Connect the oscilloscope as shown in figure 12.

c. Measure the Super Group Resync Command voltage at the TED and Alarm connector.

d. Force the TD-1236/G to resync the Groups by varying the Group Simulator data rate.

e. Measure the four Group Resync Command voltages at the TED and Alarm Connector using the oscilloscope connected as in figure 12.

f. Verify that the requirements of 3.7.2.1 are met.

4.7.2.1.1 Resync command waveform. (See 3.7.2.1)

a. Connect the equipment as shown in figure 13.

b. Set the UUT and Group Simulator as shown in 4.7.2.1.

c. Measure the Super Group Resync Command, rise and fall times, and pulse width.

d. Force the UUT to resync the groups by varying the Group Simulator data rate.

e. Measure the four Group Resync Command rise and fall times, and pulse width.

f. Verify that the requirements of 3.7.2.1 are met.

4.7.2.2 Frame squelch interface. (See 3.7.2.2)

a. Connect the equipment as shown in figure 14. Make no connections to the Group TED and ALARM connector at this time.

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b. Set the UUT as in 4.7.1.1 with the Group Simulator providing Frame Acknowledge (FRA) transmit formated with looped timing.

c. Frame the UUT by changing the Group Simulator from FRA to MSF (Multiplex Signal Format) pushing the sync button on the Group Simulator, until the unit is passing data with no errors.

d. Lower the Super Group Rate Select switch until the "OFF" LED remains lit. Then return the Super Group Rate switch to 14 and repeat Step c. The unit must frame (no errors indicated on Group Simulator).

e. Connect the DC Power Supply, via the attenuator to the TED and ALARM connect. Then set the DC level to 0.8V, and repeat Step d. Verify that the unit will not frame up (errors indicated on Group Simulator).

f. Raise the DC voltage at the TED and ALARM connector until it is 2.0 V, and repeat Step d. Verify that the unit frames up (no errors indicated on Group Simulator).

4.7.3 Multiplexing. (See 3.7.3)

4.7.3.1 Single group input tests.

a. Connect the equipment as shown in figure 15.

b. For each Group Rate tested, set the GROUP MUX RATE switch on the GROUP MUX CCA and the SUPER GROUP A RATE switch on the SUPER GROUP A CCA in the positions given. Set Timing Switch (TMG) to Station Clock and each Group Frame switch to YES.

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TABLE XIII. Control Settings on UUT for Single Group Input Tests.

GROUP MUX RATE SELECT SWITCH		SUPER GROUP A RATE SWITCH		FORMAT SWITCH	CHANNEL RATE, Kb/s
Position	Rate, kb/s	Position	Rate, kb/s		
2	128	2	128	7	16
3	144	3	144	7	16
4	256	4	256	7	16, 32
5	288	5	288	7	16, 32
6	512	6	512	7	32
7	576	7	576	7	32

c. Apply Group Traffic to Group 1 at each of the rates given. At each rate, verify that there are no errros for 10 seconds.

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d. Set UUT for operation at Group Rate 288 and Channel Rate 16 kb/s. Set the Group Simulator to Subchannel 6 and verify that there are no subchannel errors for 30 seconds.

e. Repeat step d for UUT operation at Group Rate 256 kb/s and Channel rate 32 kb/s.

f. Repeat step d and e with the Group Simulator set to Subchannel 7.

g. Connect the Group Simulator to the Super Group input and loop back Group 1. Set UUT timing to Looped and Group Simulator timing to internal.

h. With the Group Simulator set to Subchannel 5, 6, and 7, repeat steps d and e.

4.7.3.2 Rate combination tests.

a. Connect the equipment as shown in figure 16. The Group Simulators provide test signals for four group inputs.

b. For each test Rate Combination given in table I, II and III of 3.7.3 and not tested in 4.7.3.1, set the appropriate GROUP MUX RATE SELECT SWITCH on the GROUP MUX CCAs, the SUPER GROUP RATE SWITCH on the SUPER GROUP A CCA and the SUPER GROUP FORMAT SWITCH on the SUPER GROUP CCA as indicated in the table entry. Set the Channel Rate to 32 kb/s and each GROUP FRAME to YES.

c. Set the Group Simulator for LOOPED TIMING and properly synchronized for MSF operation.

d. For each rate given above, measure errors through each active group and verify that there are no errors in 10 seconds for any active group.

e. For each Group 1 measurement, measure errors on Subchannels 6 and 7. Verify that there are no errors for 30 seconds.

f. Set Group Simulator and UUT Channel Rates to 16 kb/s and repeat steps b through e.

NOTE: Group and Super Group Rates are 1/2 that shown in tables.

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4.7.4 Framing. (See 3.7.4)

4.7.4.1 Frame acquisition time. (See 3.7.4.1)

a. Connect the equipment as shown in figure 17.

b. Set the UUT as in 4.7.1.1.

Group Rate:	9 (1152 kb/s)
Frame:	YES
Timing:	STATION CLOCK
Super Group Rate:	14 (4608 kb/s)
Channel Rate:	32 kb/s
Format	2

c. Set up Error Inserter Test Set for 10^{-3} error rate and synchronize the Group Simulator so there are no errors.

d. Squelch Super Group Data at the Error Inserter Test Set to cause loss of frame. Release squelch to restore data.

e. Measure the time from resumption of data to frame-in. Use the squelch release signal at the Error Inserter Test Set to start time-interval measurements and the signal at Pin 33 of CCA A7 of the UUT to stop time-interval measurement.

f. Verify that no more than 5 of 50 measurement exceed 50 ms.

g. Set up the Error Inserter for 20% BER random bursts interspersed with 0.1% BER random error intervals.

h. Repeat e through h.

i. Verify that no more than 5 of 50 measurements exceed 100 ms.

4.7.4.2 Frame maintenance*. (See 3.7.4.2)

a. The equipment connections and the Group Simulator and UUT switch positions are the same as in test in 4.7.4.1.

b. Set the Error Inserter for continous 20% BER random errors.

c. Monitor the UUT for 12 hours. Verify that there are no drop-outs during the 12 hour period.

*NOTE: Accelerated Error Environment.

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4.7.4.3 Detection of out-of-sync. (See 3.7.4.3)

a. The equipment test setup and control settings are the same as in figure 17.

b. Set up Error Inserter Test Set for no errors.

c. For the 1152 kb/s Group Rate perform d and e to determine the frame drop-out time of the UUT.

d. Squelch Super Group data at Error Inserter Test Set. Measure time from traffic interruption to out-of-frame indication.

e. Use the data squelch signal at the Error Inserter Test Set to start time-interval measurement. Use the signal to pin 33 of CCA A7 of the UUT to stop time-interval measurement. Take 50 measurements.

f. Verify that the requirements of 3.7.4.3 are met.

4.7.4.4 Delayed frame search. (See 3.7.4.4)

a. The equipment test setup and control settings are the same as in figure 17.

b. Use a Pulse Generator to inhibit the Super Group data at the Error Inserter Test Set. Determine that the single-shot pulse is 450 ms.

c. Squelch Super Group data at the Error Inserter Test Set for 450 ms by using Pulse Generator connected to the Data Inhibit connector of the Error Inserter Test Set.

d. Verify that the out-of-frame indication of the SUPER GROUP A CCA is not indicated.

4.7.4.4.1 False frame lock.

a. Connect the equipment as shown in figure 18.

b. Set the UUT as follows:

Group Rate:	3 (144 kb/s)
Frame	YES
Timing:	STATION CLOCK
Super Group Rate:	5 (288 kb/s)
Channel Rate:	32 kb/s
Format:	6

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c. Set the TD-1233 to 32 kb/s Channel Rate, and the MD-1026 to Channel Rate 3 and MOW.

d. Monitor the DSVT Simulator. Verify the presence of activity with no errors.

e. Turn the TD-1233 ON and OFF 50 times. Verify that channel activity with no errors occurs each time the TD-1233 is turned ON.

4.7.4.5 Idle filler pattern. (See 3.7.4.5)

a. Connect the equipment as shown in figure 19.

b. Set the UUT as follows and the Group Simulator to provide a NO Frame transmit format.

Group Rate:	9 (1152 kb/s)
Frame:	NO
Timing:	STATION CLOCK
Super Group Rate:	14 (4608 kb/s)
Channel Rate:	32 kb/s
Format:	2

c. Observe the data signals out of the UUT at pins E and F on the Group 1 connector.

d. Verify that the idle filler pattern is as specified in 3.7.4.5.

NOTE: All data channels shall be ones except for the framing bit.

e. Repeat c and d for Group 2, Group 3, Group 4, and Super Group port on the UUT.

NOTE: When making the Super Group measurement, insert the Group Simulator signal into the SUPER GROUP connector directly, and adjust the bit rate to the Super Group Rate of 4608 kb/s. Observe the Groups 1 to 4 and Super Group outputs.

4.7.4.6 Group input buffering. (See 3.7.4.6)

a. Connect the equipment as shown in figure 20 with the Timing Source locked to the synthesizer internal time base.

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b. Set the Group Simulator to NO Frame transit format and set the UUT as follows:

Group Rate:	5 (288 kb/s)
Frame:	NO
Timing:	Loop
Super Group Rate:	9 (1152 kb/s)
Channel Rate:	32 kb/s
Format:	2

c. With the Frequency Synthesizer, set the Group Input to 288,003 (+3 Hz over nominal).

d. Press the ERROR CHECK SYNC button on the Group Simulator. At the error indication, reset the sync and measure the time interval to the subsequent error burst. The interval shall be greater than 10 sec.

e. Set the Frequency Synthesizer to 287,977 (-3 Hz under nominal).

f. Repeat d.

g. Verify that the requirements of 3.7.4.6 are met.

4.7.5 Prime power. (See 3.7.5)

a. Connect the equipment as shown in figure 21.

b. Set the Group Simulator as follows:

Group Rate:	9 (1152 kb/s)	Type: 1
Timing Mode:	LOOP	Channel Rate: 32 kb/s
Transmit Format:	MSF	Data Select: Random

Set the UUT as follows:

Group Rate:	9 (1152 kb/s)	Super Group Rate: 14 (4608 kb/s)
Frame:	YES	Channel Rate: 32 kb/s
Timing:	Station Clock	Format: 2

c. Connect the AC power source to the POWER Connector on the UUT.

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d. For each of the combinations of voltage and frequency given in 3.7.5, perform the following measurements:

(1) Measure the Input Power.

(2) Measure the group BER at 1152 kb/s.

(3) For each of the combinations of voltage and frequency given in 3.7.5, verify that the input power is 75 watts maximum and no group errors over a 10 second period.

4.7.6 Built-in test equipment (BITE) and alarms. (See 3.7.6)

4.7.6.1 No fault operation.

a. Connect the equipment as shown in figure 13 using ohmmeters to monitor the S and SA Alarms at the TED and Alarms connectors.

b. Set UUT as in 4.7.5.

c. Verify that the BITE requirements listed below are met.

(1) S Open

(2) SA Open

(3) LOI and SUMMARY FAULT lamp must be OFF.

(4) POWER SUPPLY OUT OF TOLERANCE (POT) lamp must be OFF.

(5) All CCA LEDs must be OFF.

4.7.6.2 Summary fault.

a. Press the Lamp Test switch on the UUT.

b. Verify that the BITE requirements listed below are met.

(1) S must go to closed.

(2) SA must go to closed.

(3) LOI, POT & SUMMARY FAULT lamp must light.

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4.7.6.3 Power loss.

a. Disconnect the power at the UUT source with the UUT power switch set to ON.

b. Verify that the BITE requirements listed below are met.

(1) S must go to closed.

(2) SA must go to closed.

4.7.6.4 Power Off.

a. Turn OFF the UUT power switch.

b. Verify that the BITE requirements listed below are met.

(1) S must go to closed.

(2) SA must go to open.

4.7.6.5 BITE CCA removal.

a. Remove the BITE CCA.

b. Verify that there are no errors on the Group Simulator for 10 seconds.

4.7.7 Operational check. The equipment setup and UUT and Group Simulator switch settings are the same as in 4.7.6.

4.7.7.1 Group data error check. (See 3.7.7) Monitor the Group Error Detector set at 1152 kb/s for a period of 10 seconds. Verify that there are no errors.

4.7.7.2 Alarm monitor check. Verify that there are NO ALARMS generated.

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4.7.7.3 Telemetry error check.

a. Set the Group Simulator to Subchannels 6 and 7. At each setting, monitor the subchannel for 30 seconds.

b. Verify that there are no errors in either subchannel.

4.7.8 Timing modes. The Timing Modes Tests are performed as part of test in 4.7.3.

4.7.9 Environmental full duplex signal detection test. (See 3.7.9)

a. Connect the equipment as shown in figure 22.

b. Set the UUT as follows:

Group Rate:	9
Frame:	YES
Super Group Rate:	14
Channel Rate:	32 kb/s
Format:	2
Timing:	Loop

c. Set the AUX TD-1236 as follows:

Group Rate:	9
Frame:	YES
Timing:	Station Clock
Super Group Rate:	14
Channel Rate:	32 kb/s
Format:	2

4.7.9.1 Group data error rate check. (See 3.7.9.1) Monitor the Group Simulator for a period of 10 seconds. Verify that no errors are obtained.

4.7.9.2 Telemetry error check. (See 3.7.9.2)

a. Monitor errors on Subchannel 6 on the Group Simulator for 30 seconds.

b. Verify that no errors are obtained.

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4.7.9.3 Alarm monitor check. (See 3.7.9.3)

a. Monitor the S, LOI, and SA alarms. Verify that there are no alarms generated.

b. Shut OFF the power on the Auxiliary TGM and verify the S and LOI Alarms and no SA Alarm on the UUT.

4.7.9.4 Sync/frame check. (See 3.7.9.4)

a. Monitor the Super Group Resync Command signal at the TED and Alarm Connector on the UUT. Disconnect the Super Group to cable from the Auxiliary Unit. Connect it to the Group Simulator (Rate 14) and set the simulator to the NO FRAME position. The UUT shall generate a Resync Command Pulse (10 us) approximately every second.

b. Connect the test setup as before and monitor the Super Group Resync Command output. The Resync Command Pulse shall stop being generated.

c. Monitor the Group 1 Resync Command signal at the TED and Alarm connector on the UUT. Turn OFF the Auxiliary Unit Power. The unit shall generate a Resync Command Pulse.

d. Turn ON the Auxiliary Unit Power. The unit shall stop generating a Resync Command Pulse.

e. Repeat steps c and d for Groups 2, 3 and 4.

4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2 Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +63°C (+145°F). Maintain internal chamber temperature for 6 hours at +63°C (+145°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +63°C (+145°F) within a time period of one hour. Repeat the above steps two additional times for a total of three 12-hours cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +63°C (+145°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

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4.8.1.2 Low temperature. (See 3.8.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C(-67°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation. (Para. II- 3.2 of Method 502.2). The chamber temperature in step 1 of Procedure II shall be adjusted to -31.7°C(-25°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III.

4.8.3 Low pressure (altitude). (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In step 2 and step 5, the air velocity shall be 200 ±100 feet per minute. In step 5, the air temperature shall be adjusted to +63°(+145°F). The equipment shall be reoriented periodically to expose all its sides to the dust stream. The equipment shall be operated before step 1, during step 8 and during step 11.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C(+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall be no clogging or binding of moving mechanical parts.

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4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs, etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3, Procedure I, Category 8 (Ground Mobile). The vibration level shall be in accordance with figures 514.3-7 through 514.3-9 of MIL-STD-810D. The axes designated on the schedules are the vehicle axes. The orientation of the equipment relative to the vehicle, therefore, must be considered for proper direction of vibration input. Test time shall be 4 hours in each of three mutually perpendicular directions for a total of 12 hours. Prior to and following these tests, the equipment shall be operated in accordance with MIL-STD-810D general test performance guidance paragraph 4.5. The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

4.8.7.2 Shock.

4.8.7.2.1 Functional. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure I. The shock pulse shall be 20g peak for a nominal duration of 11 msec in accordance with figure 516.3-4.

4.8.7.2.2 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

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4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article reliability test. First article reliability testing shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.2 Quality conformance (group D) reliability test. The reliability test for group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. The Multiplexer TD-1236/G shall be operated back-to-back.

b. For duty cycle, apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24 hour cycle.

c. Chamber temperature shall be 40°± 5°C (95°F to 113°F).

d. Vibration shall be 2.2g ±10% peak acceleration value at any nonresonant frequency between 20 and 60 CPS measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

4.10.4 Reliability test method. The detailed test procedures and conditions for performing the test outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor prepared, Government approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.7.

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4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781C. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic units for mating with case.

e. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12.

4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (Systems Safety Engineering) which can be determined visually.

4.13.1 System safety design verification check list. An inspection shall be performed on one unit only, during first article testing, utilizing SEL form 1183, to verify compliance with 3.14 (System Safety Engineering).

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4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14 2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table XIV, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

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TABLE XIV. Packing inspection provisions.

NO.	CHARACTERISTICS	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitation	Weigh & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.15 Serial numbers. (See 3.5.1) The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and CCA. The numbers shall be consecutively assigned.

4.16 Nuclear tests. (See 3.16) Applicable tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

4.17 TEMPEST tests. (See 3.17) TEMPEST tests shall be performed on the equipment in accordance with CR-CX-0174-001.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I through VII coded as follows:

5.1.1 Level A. 3Q-1-00-00-NR-X-ED

5.1.2 Level B. 2M-1-00-00-NR-X-ED

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

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6. NOTES

6.1 Intended use. The TD-1236/G is a shelter mounted unit which combines 2, 3, or 4 TRI-TAC groups (72 kb/s to 2,304 kb/s) into a Super Group at rates from 144 kb/s to 4,608 kb/s. It interfaces directly with modems and other multiplexers in the DGM family and with the Trunk Encryption Device (TED). The unit shall be suitable for operation and installation in shelters and vans.

6.1.1 Function description. The Multiplexer TD-1236/G shall synchronously multiplex/demultiplex up to four groups (72 to 2304 kb/s into/from a Super Group bit stream (128 to 4608 kb/s) and also shall provide a single channel operation (group rate equal to Super Group rate). The first channel of the Super Group shall be allocated to primary overhead information. The overhead channel shall be divided into eight subchannels and shall include framing and telemetry information. The group and Super Group input/output shall be balanced non-return-to-zero (NRZ) data and timing and shall be compatible with the KG-81 Trunk Encryption Device (TED); TD-1235/TTC Multiplexer; MD-1026/G Digital Data Modem; TD-1237/G Multiplexer; MD-1025/G Cable Driver Modem; and another TD-1236/G Multiplexer; and the balanced side interface of all members of the DGM family of equipments. The 2 kb/s telemetry channel shall be looped back internally. The multiplexer accepts on its group side from two to four balanced non-returm-to-zero (NRZ) group data square wave and timing signals at bit rates from 72 to 2304 kb/s. These are multiplexed into a Super Group at bite rates ranging from 288 to 4608 kb/s. The Super Group is put out as balanced NRZ data and square wave timing. In the opposite direction, a Super Group is accepted in the same form and demultiplexed, and four groups are supplied in the same form on the group side. The multiplexer shall also accept a single group on its group side and put out a reframed subgroup at the same bit rate. A similar operation is performed in the opposite direction. This single group operation shall be provided from 28 to 128 kb/s. The multiplexer also accepts a station clock signal as a balanced square wave. The frequency of the clock shall be equal to the Super Group bit rate.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

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b. Level A and B of preservation packing. (See Section 5)

c. When first article rough handling tests are not required.

d. Quantity of first article inspection units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirement.

i. MIL-Q-9858.

j. Length of cables (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. (See 4.5.3.3) Initial shipments of end items will be contingent upon successful completion of all first article tests.

6.5 Location of operational inspection. (See 4.7.7) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table XI.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject terms (key-words) list.

Balanced Equipment-side Signal
Digital Group Multiplex
Group
Multiplex/Demultiplex
Multiplexer
Multiplexer, TD-1236/G
Resync Command
Subchannels
Supergroup
Synchronous
Telemetry
Trunk Group Multiplexer (TGM)

CUSTODIAN:
ARMY-CR

Preparing Activity
ARMY-CR
Project Nr. 5820-A832

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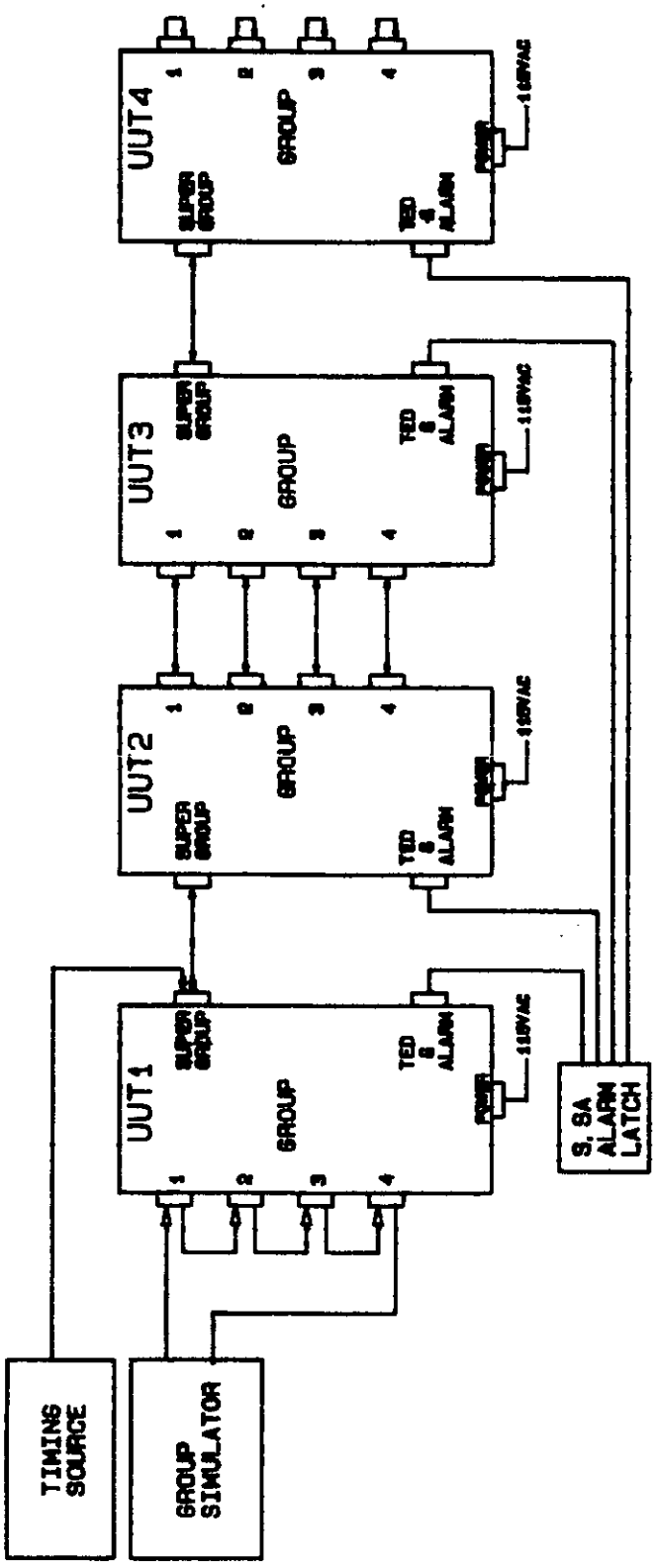
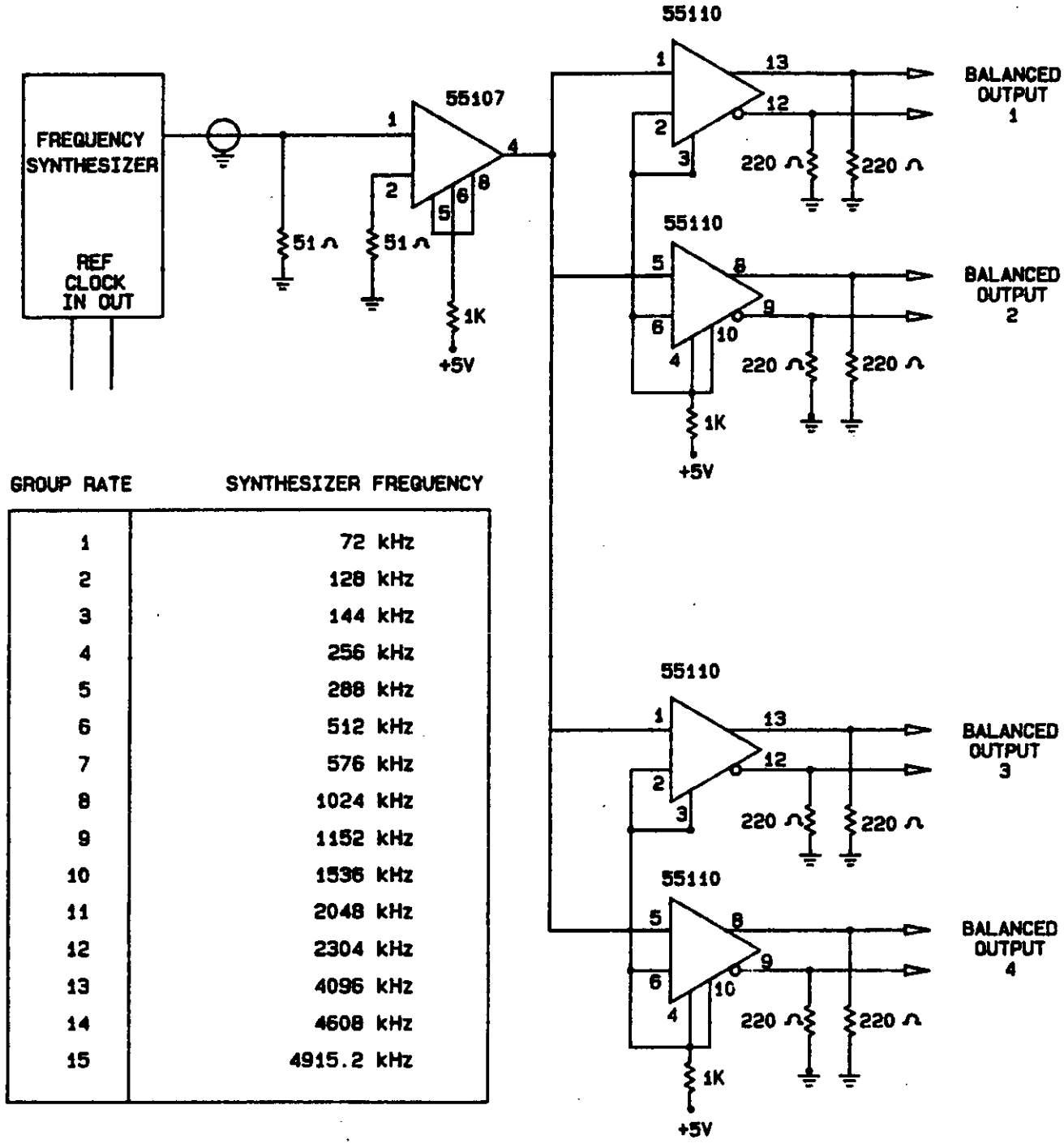


FIGURE 1. Burn-in.

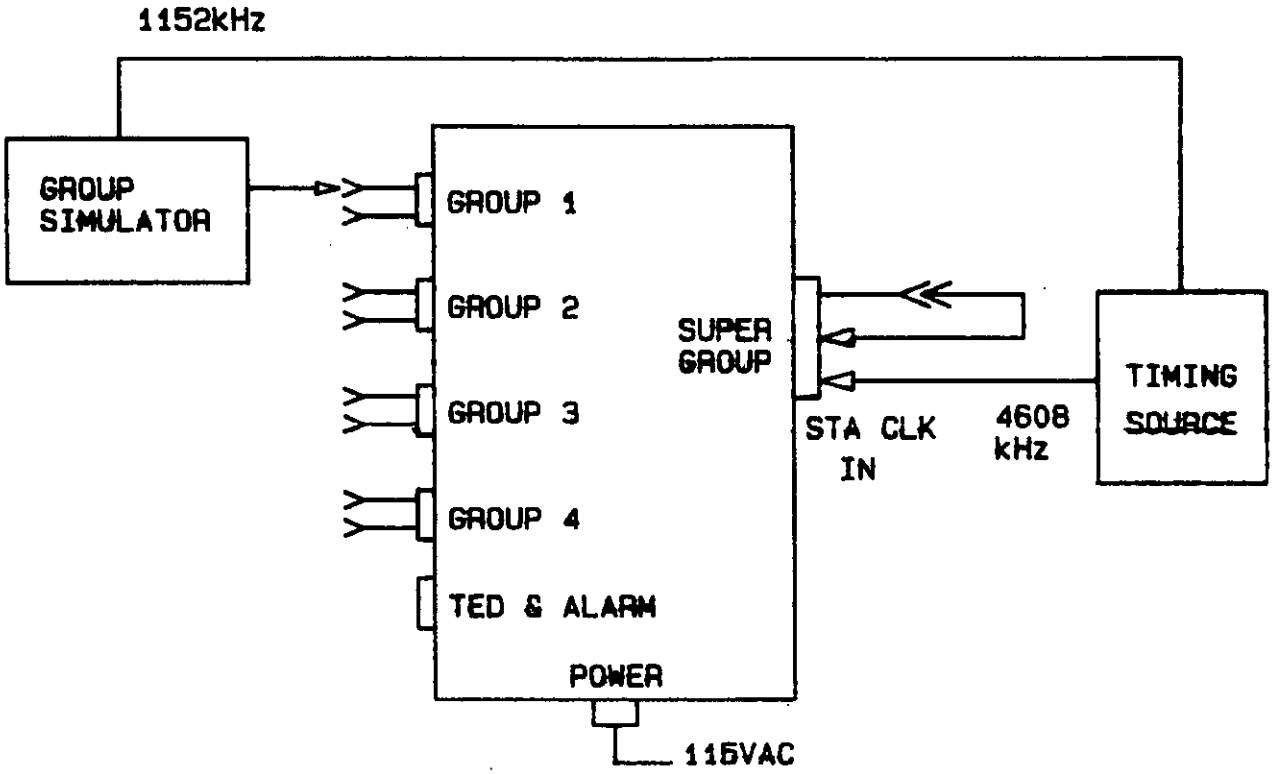
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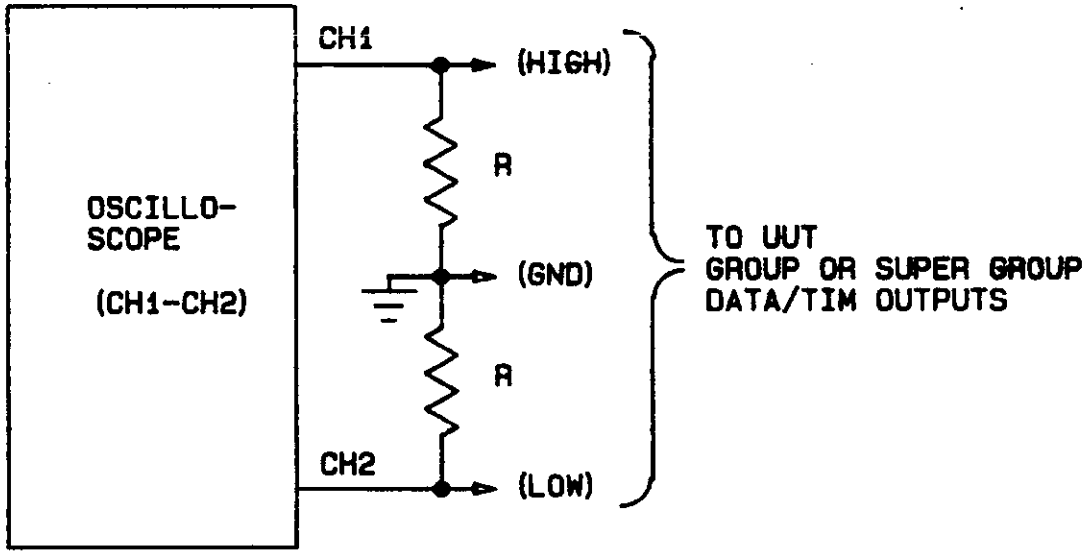


GROUP RATE	SYNTHESIZER FREQUENCY
1	72 kHz
2	128 kHz
3	144 kHz
4	256 kHz
5	288 kHz
6	512 kHz
7	576 kHz
8	1024 kHz
9	1152 kHz
10	1536 kHz
11	2048 kHz
12	2304 kHz
13	4096 kHz
14	4608 kHz
15	4915.2 kHz

FIGURE 2. Alternate timing source.

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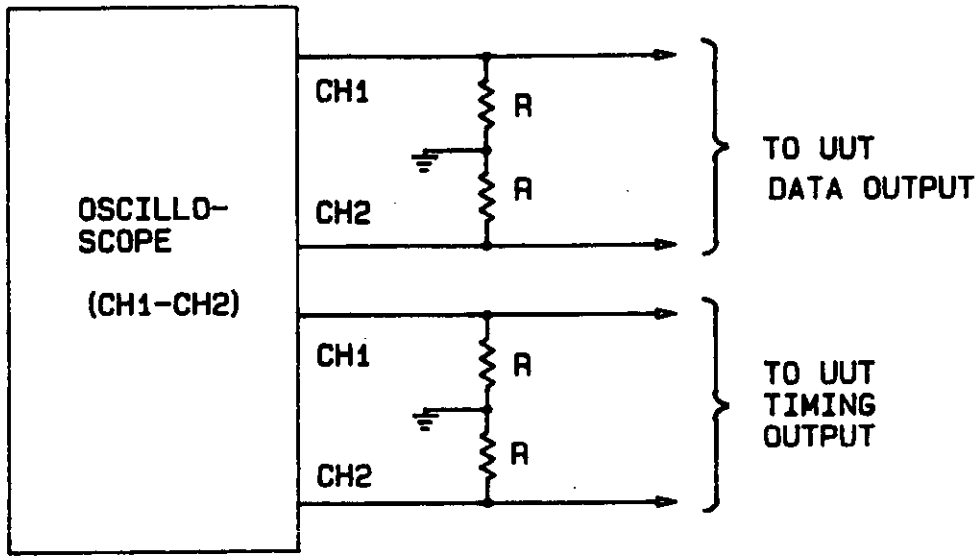




R 50 OHMS ±1%

FIGURE 3. Line-to-line voltage levels.

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FIGURE 4. Clock/data phasing.

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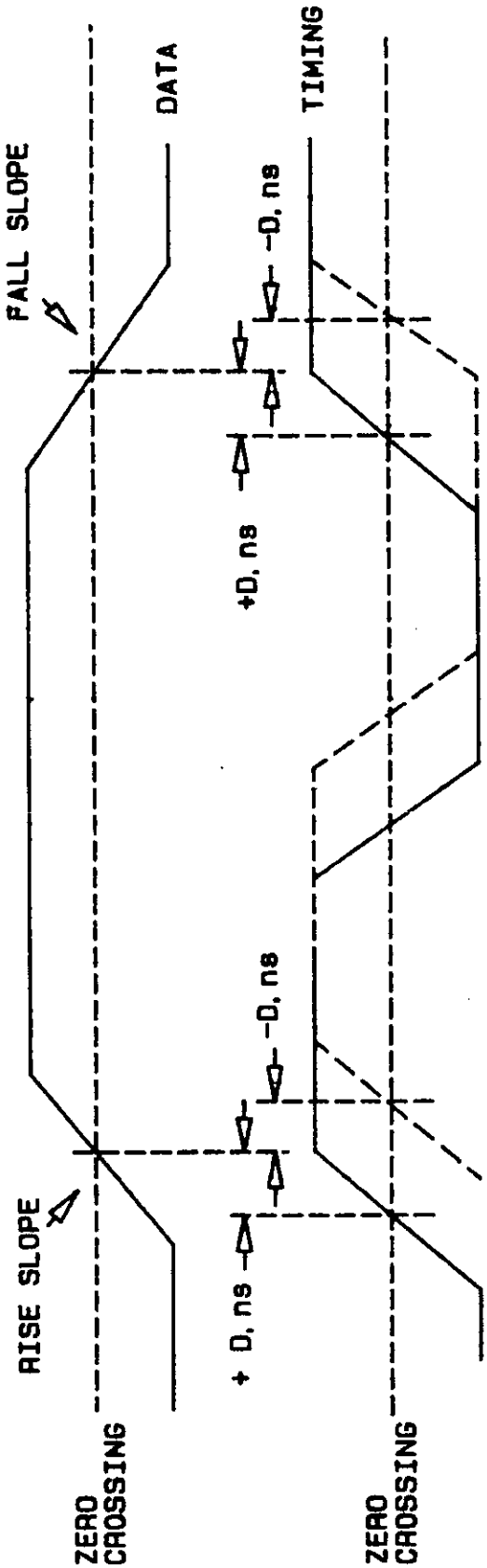
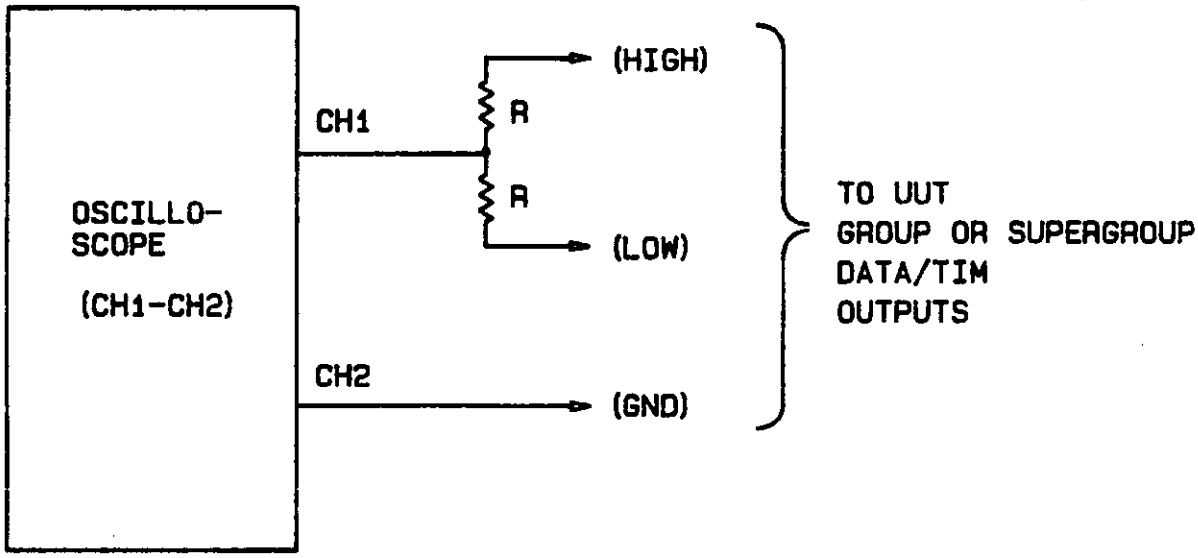


FIGURE 5. Delay measurement (clock/data phasing).

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R 50 OHMS ±1%

FIGURE 6. Offset voltage.

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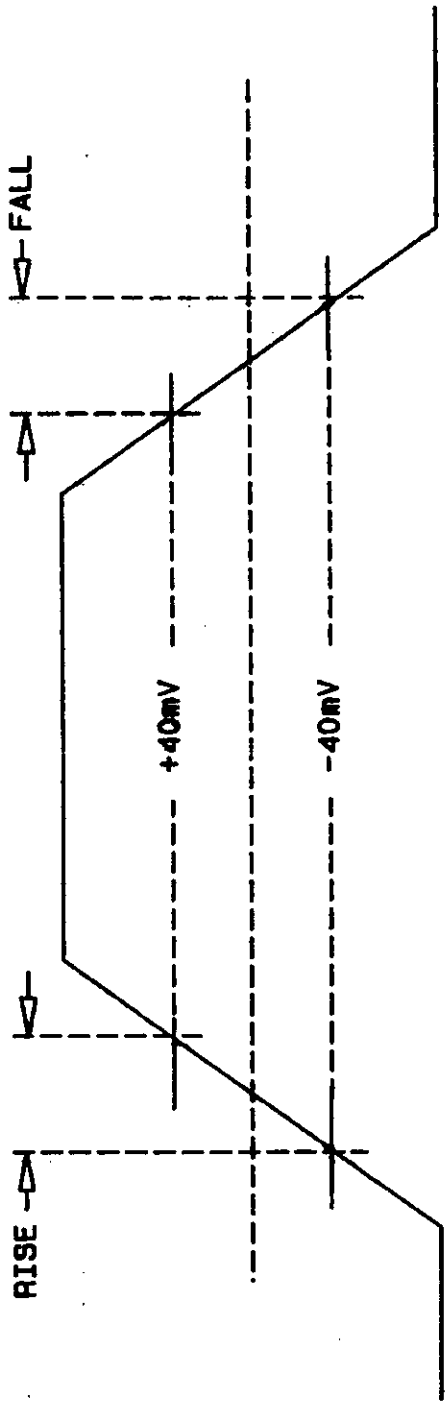


FIGURE 7. Rise and fall slopes (slew rate).

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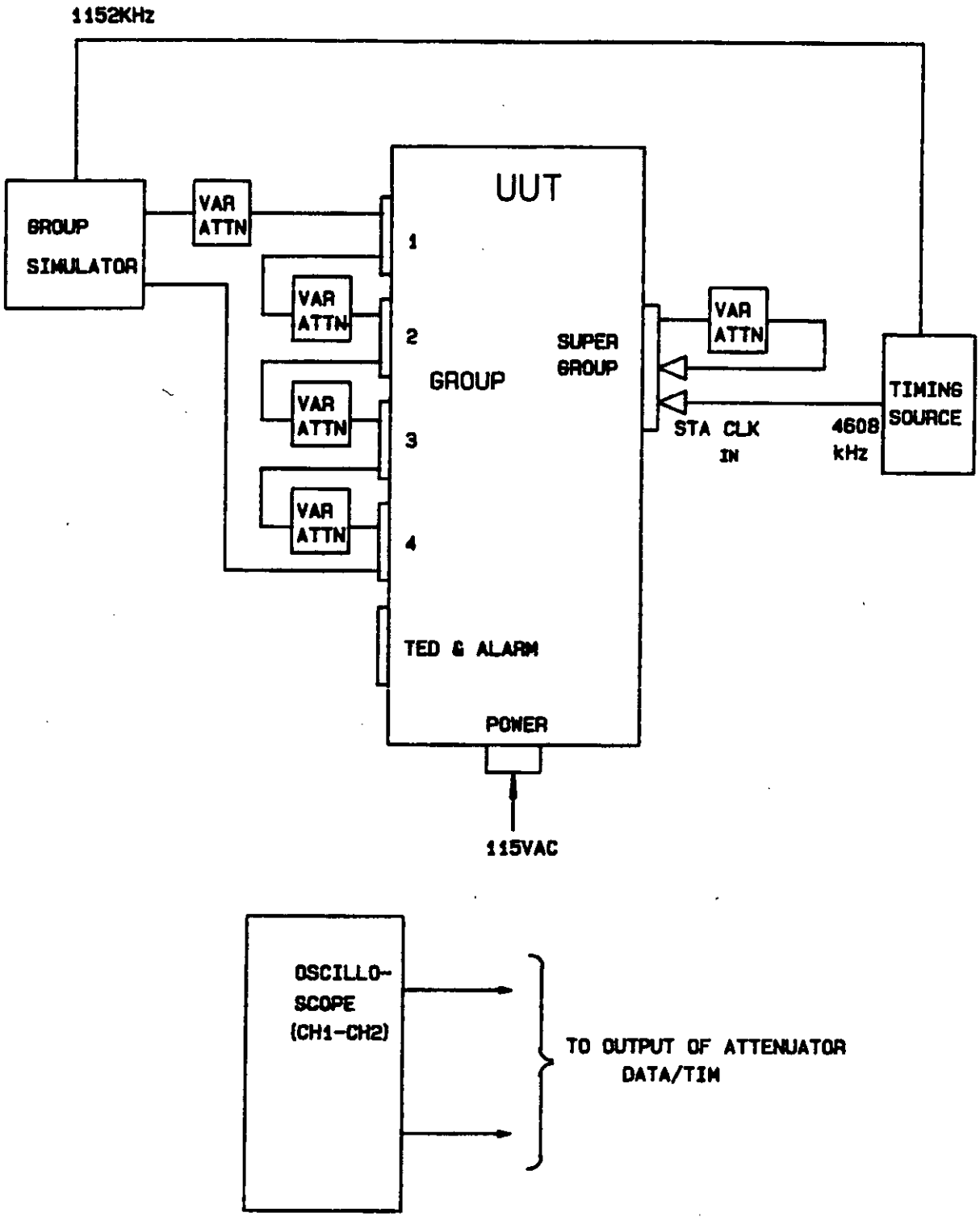
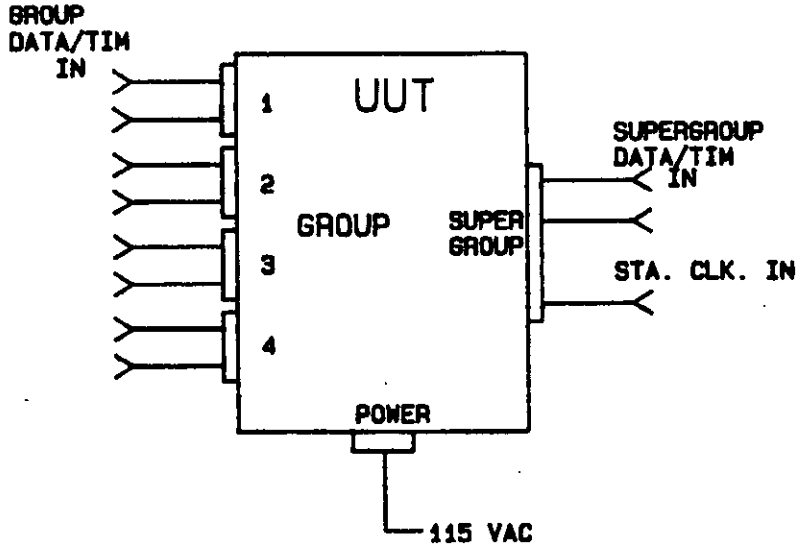


FIGURE 8. Receiver sensitivity.

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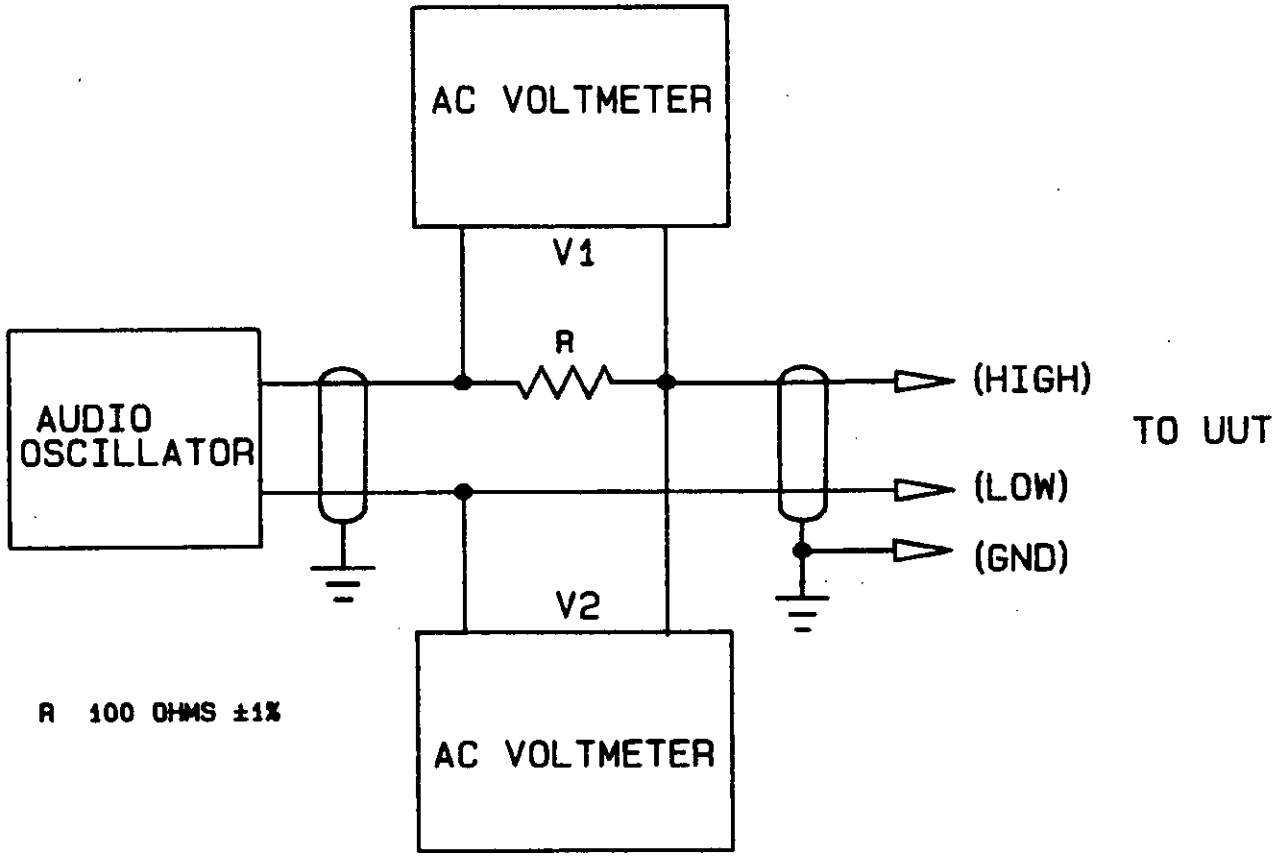


FIGURE 9. Input load impedance.

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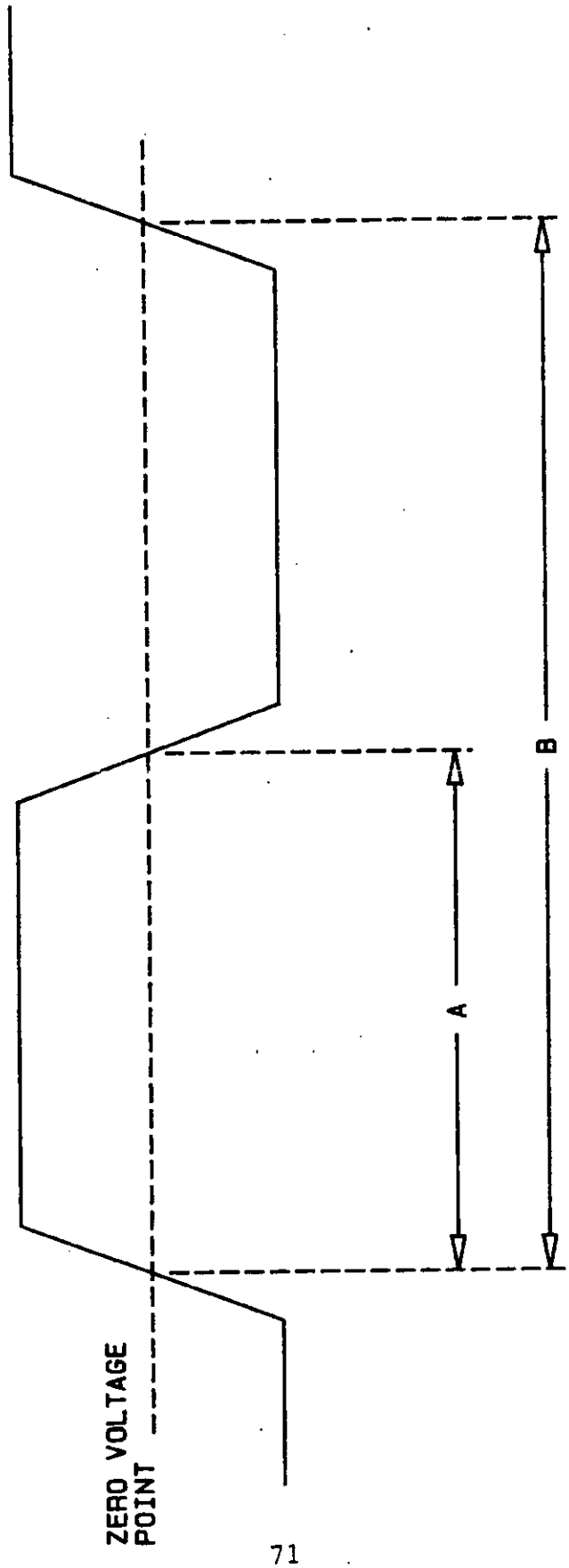


FIGURE 10. Clock asymmetry.

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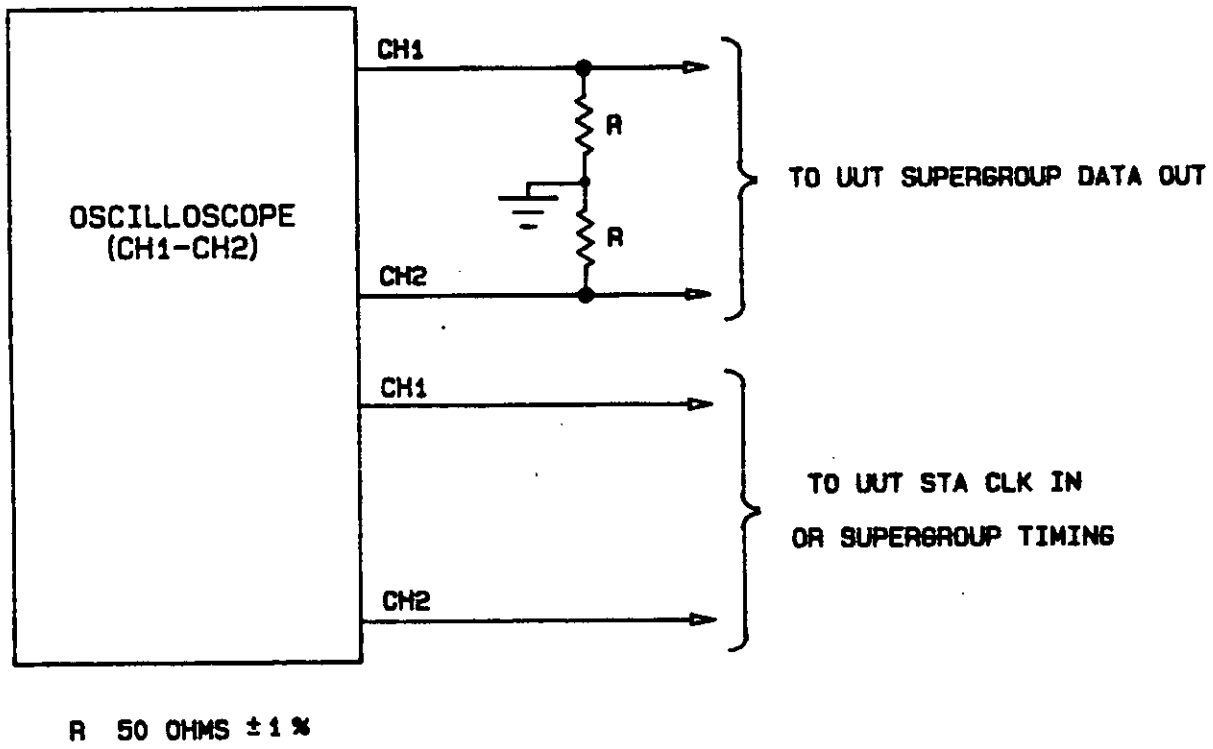
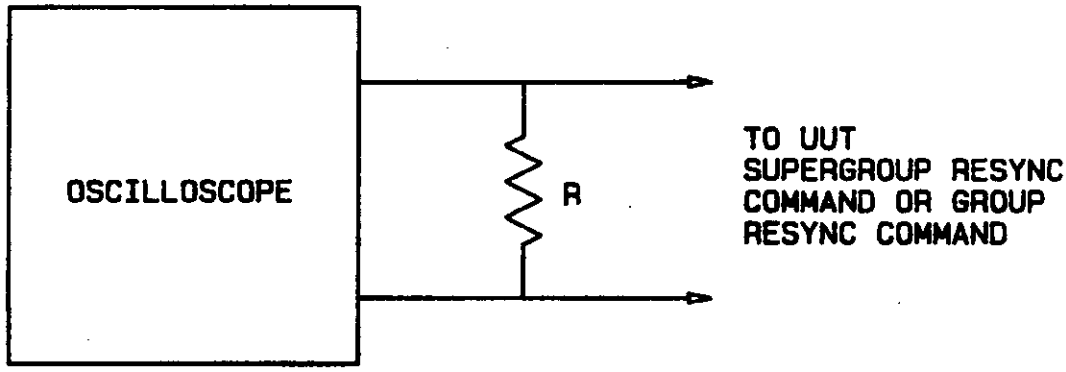


FIGURE 11. External clock/data phasing.

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R 6K OHMS ± 1 %

FIGURE 12. Resync command voltage; waveform.

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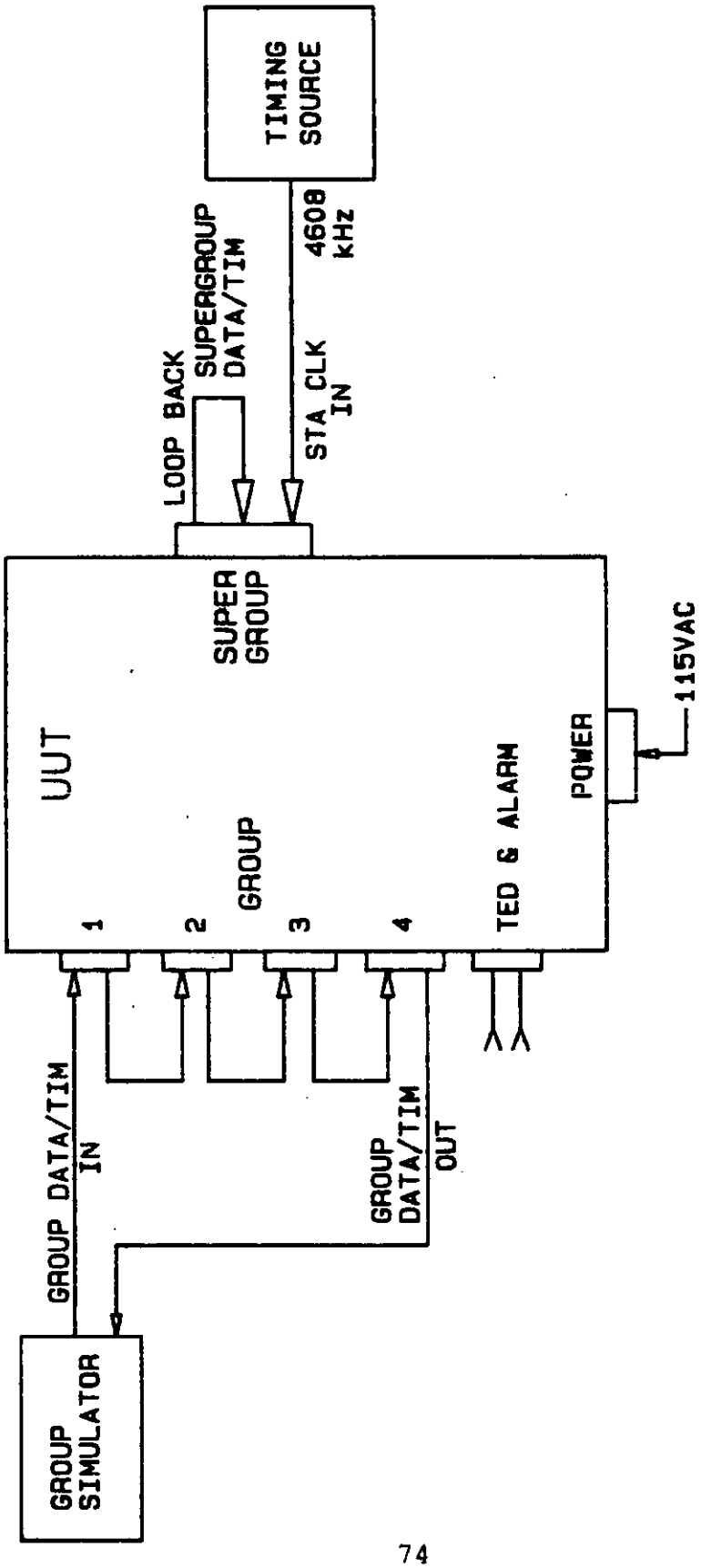


FIGURE 13. Group resync command voltage; waveform.

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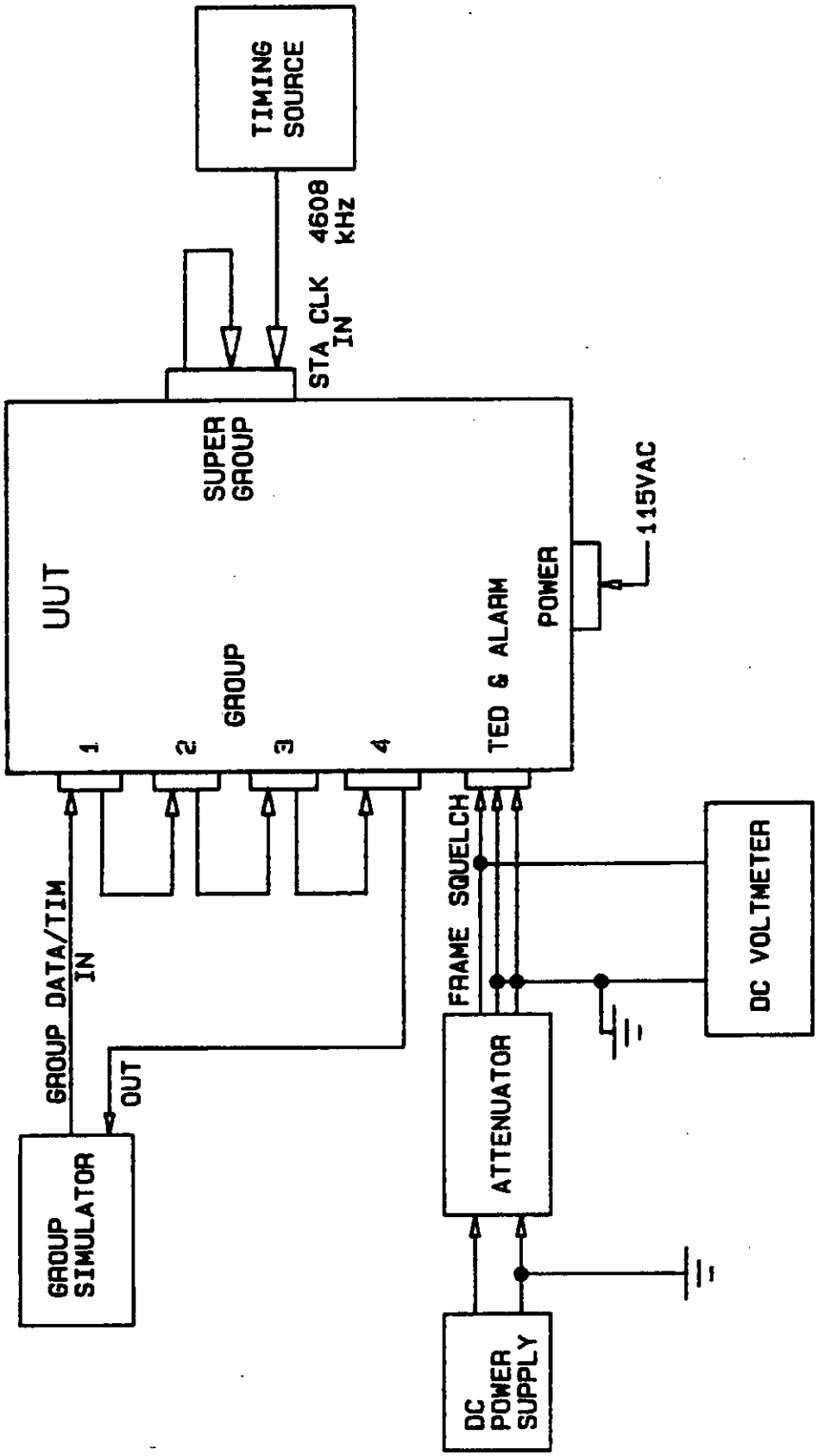


FIGURE 14. Frame squelch.

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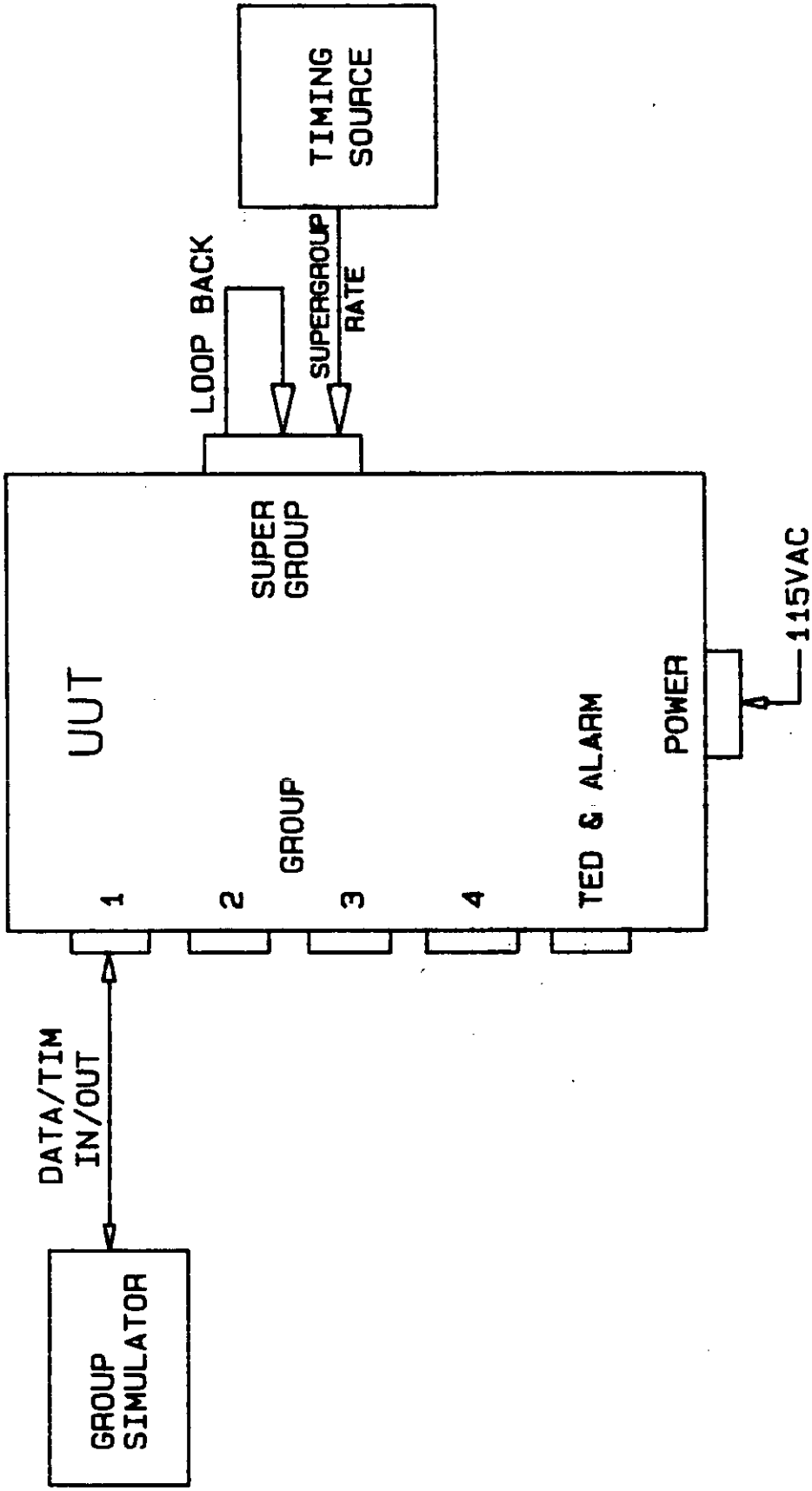


FIGURE 15. Multiplexing: single group input.

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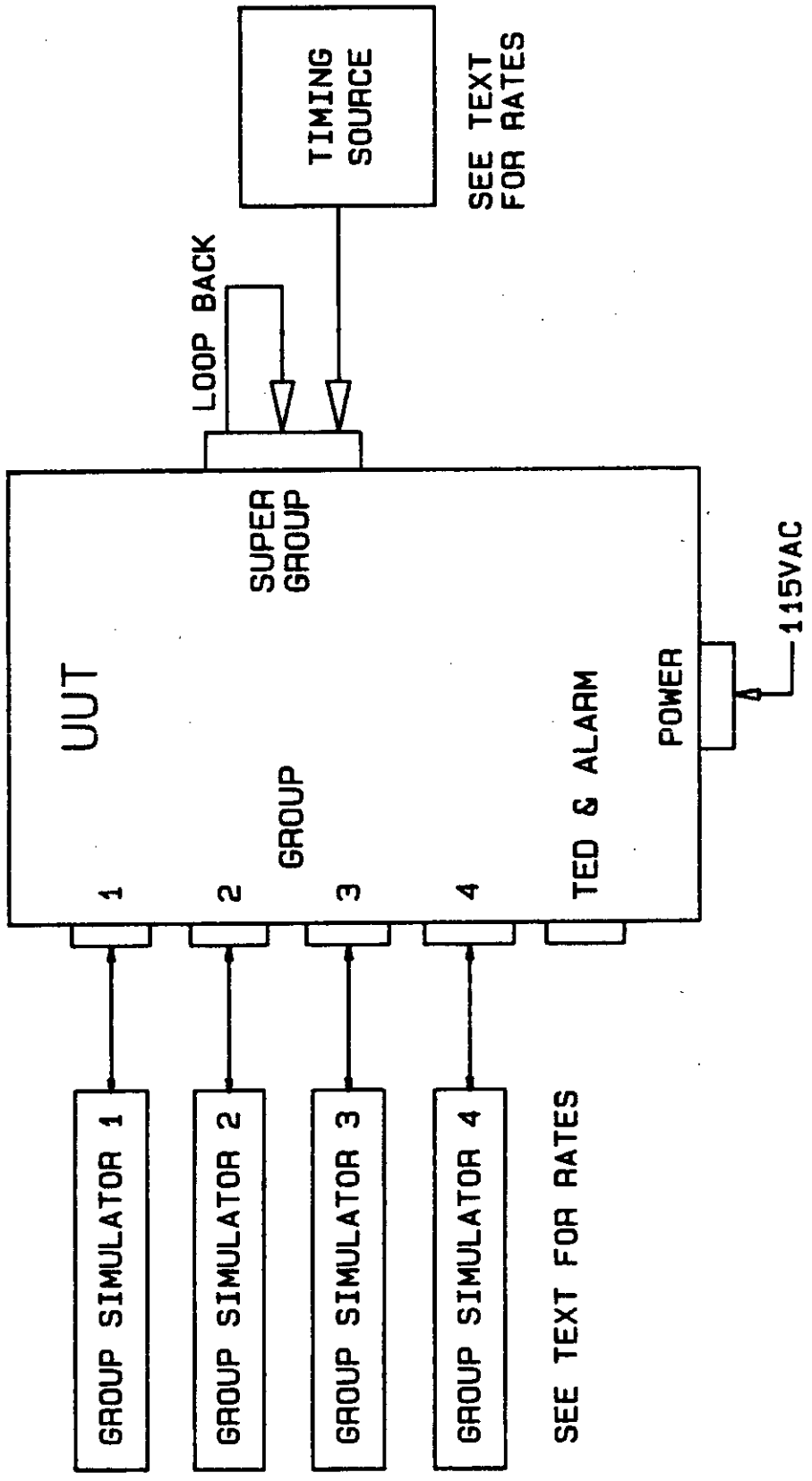


FIGURE 16. Multiplexing: rate combination.

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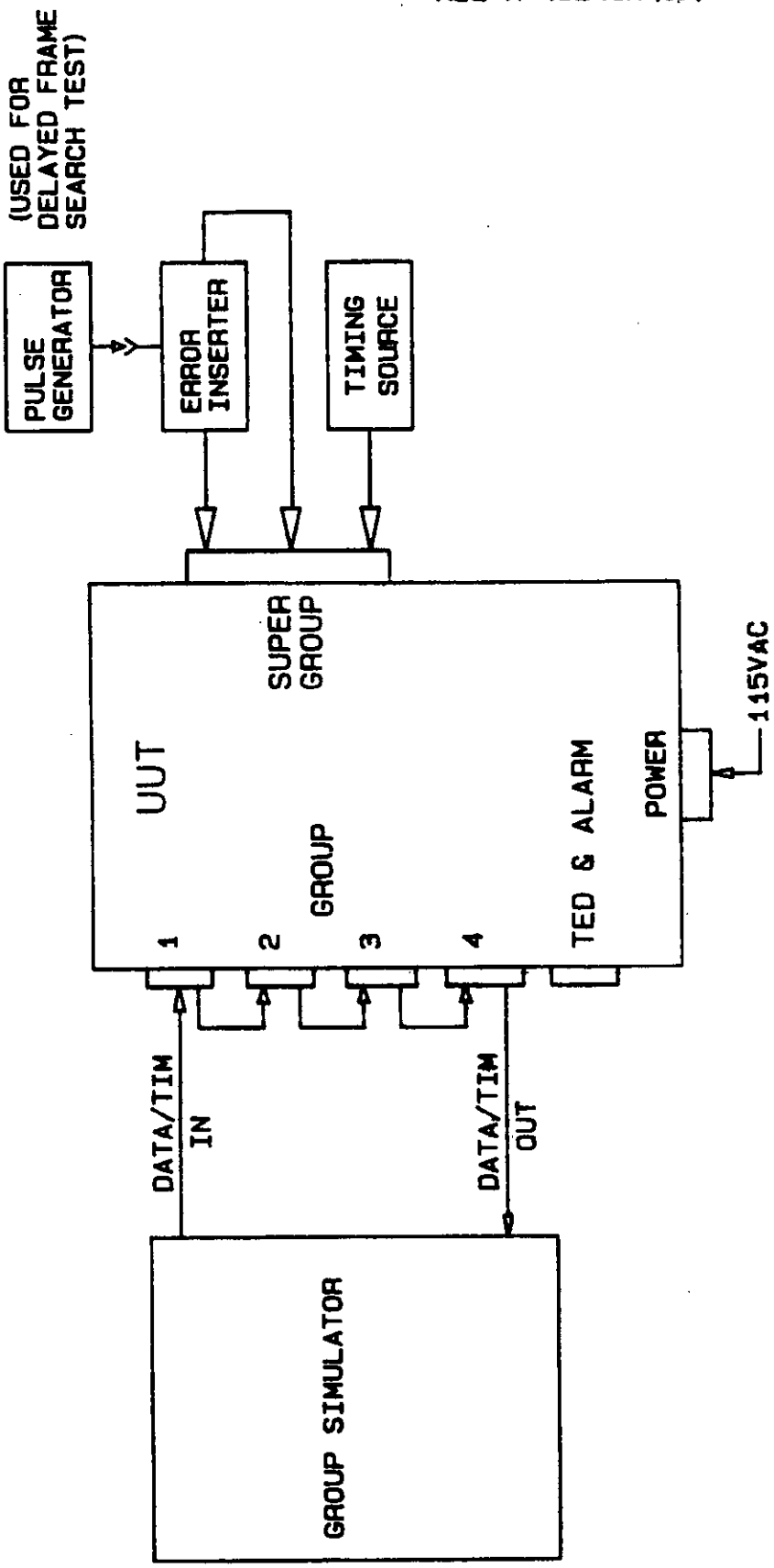


FIGURE 17. Framing test.

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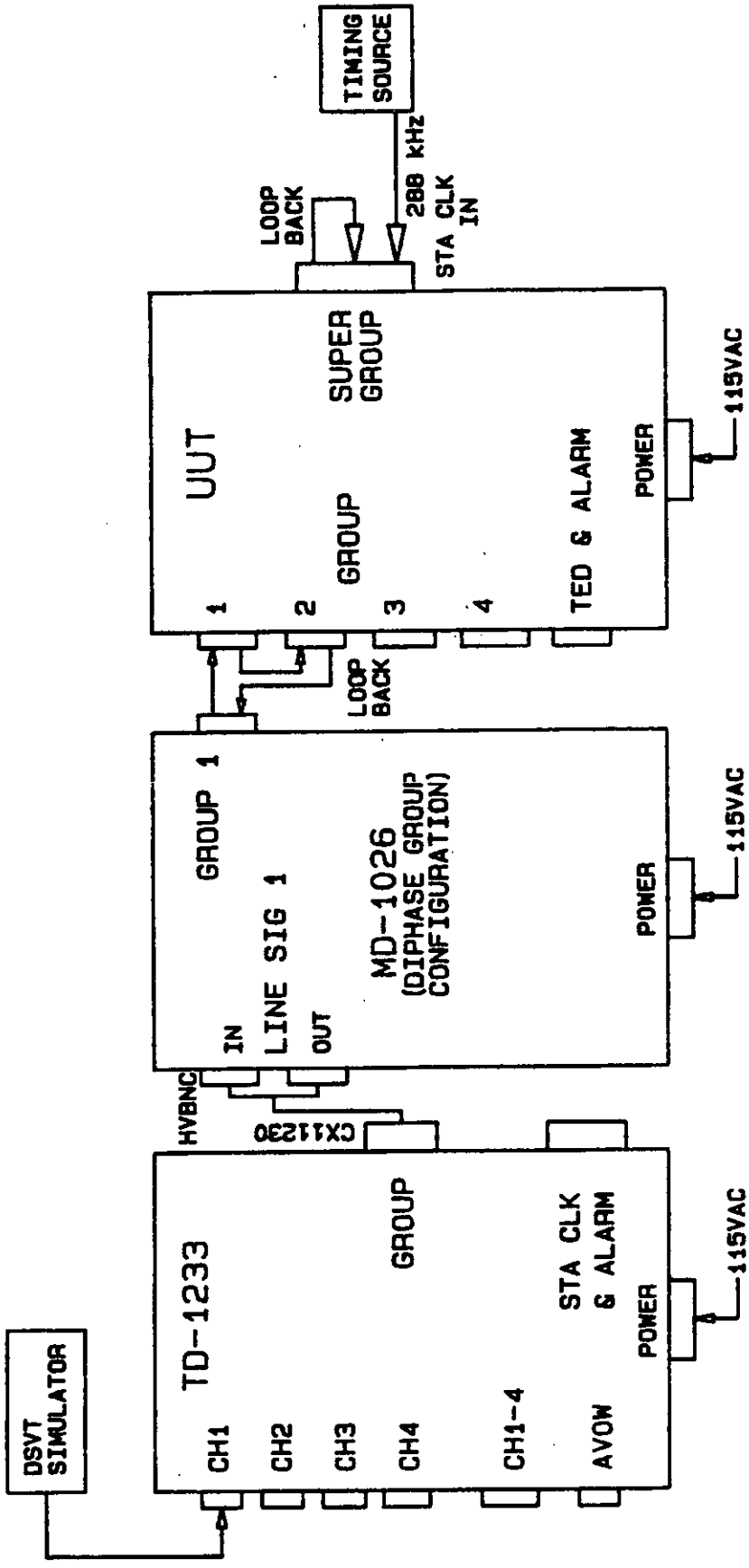


FIGURE 18. False frame lock.

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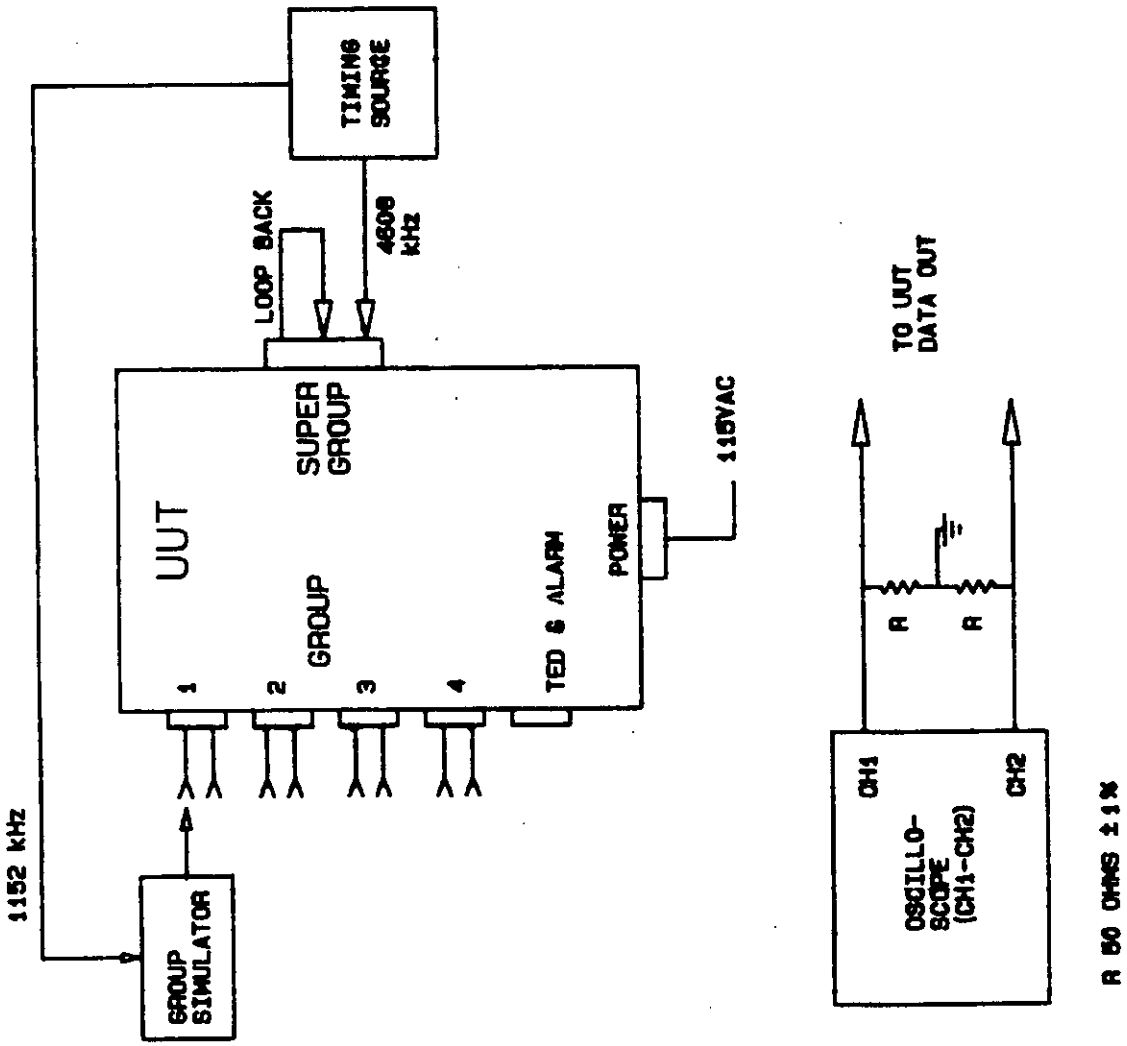


FIGURE 19. Idler filler pattern.

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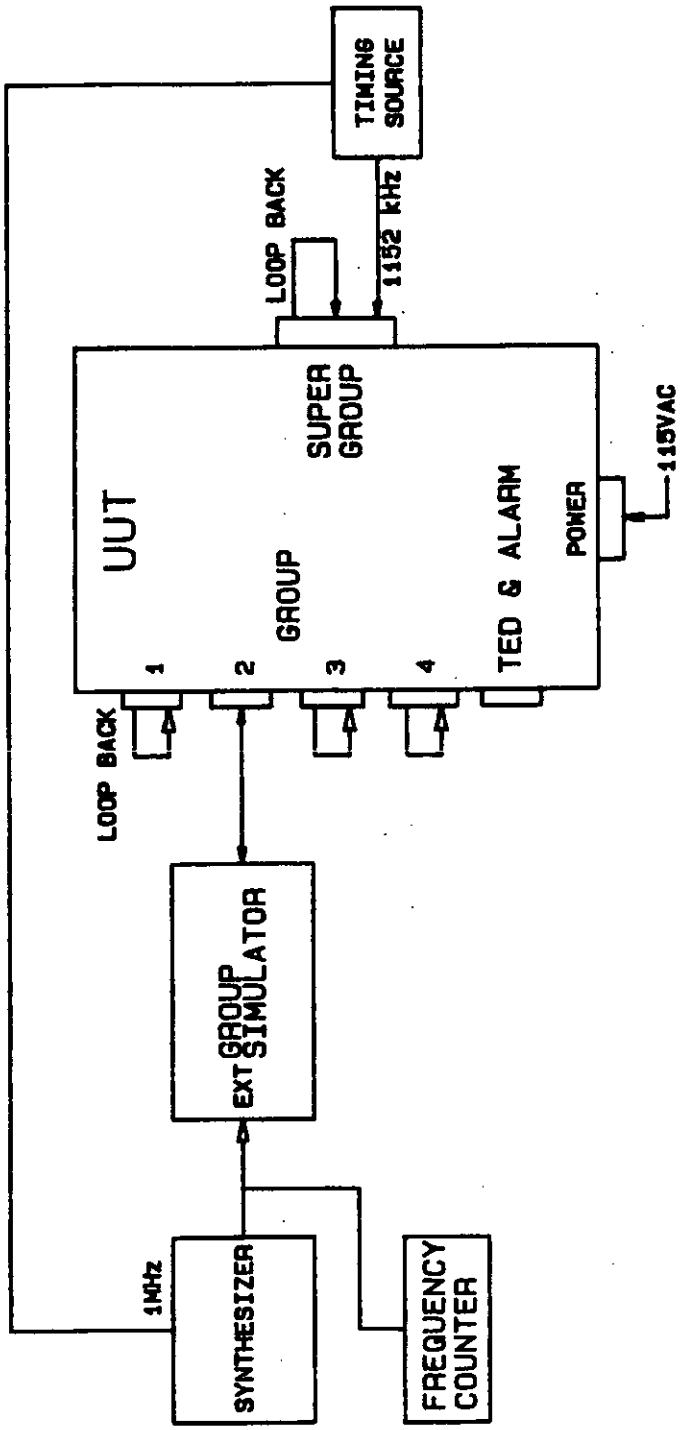


FIGURE 20. Buffer overflow.

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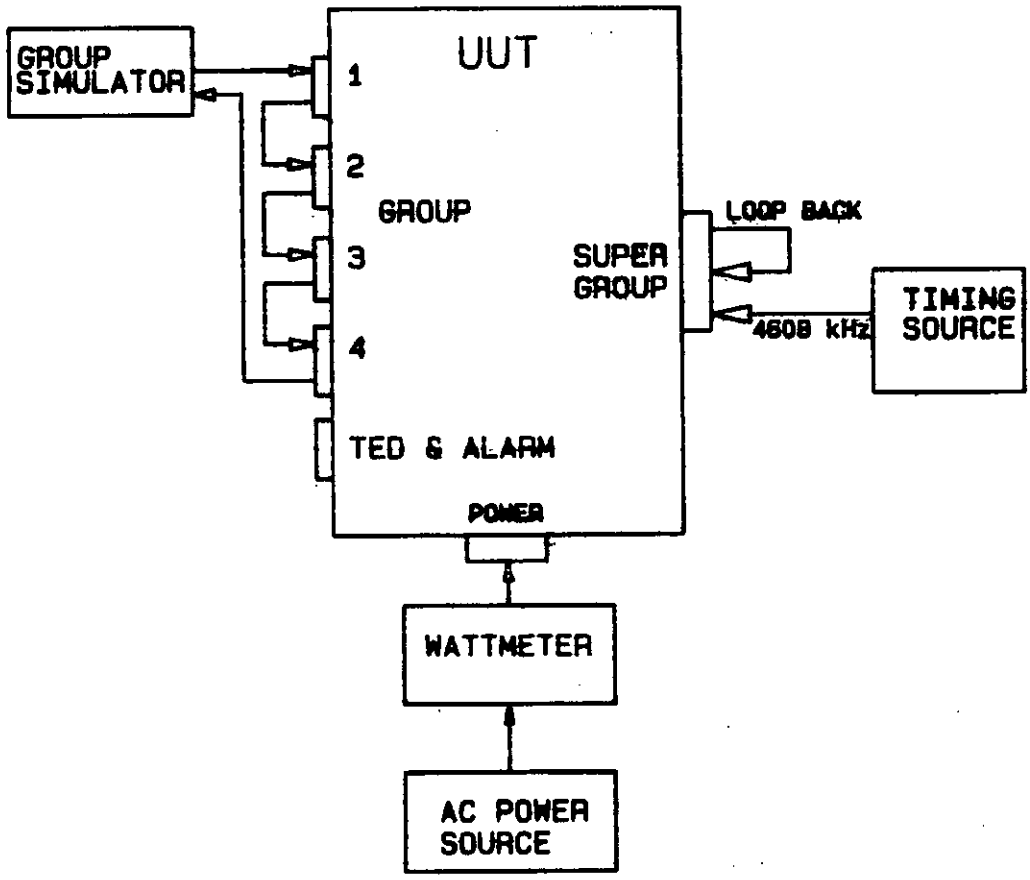


FIGURE 21. Power variation.

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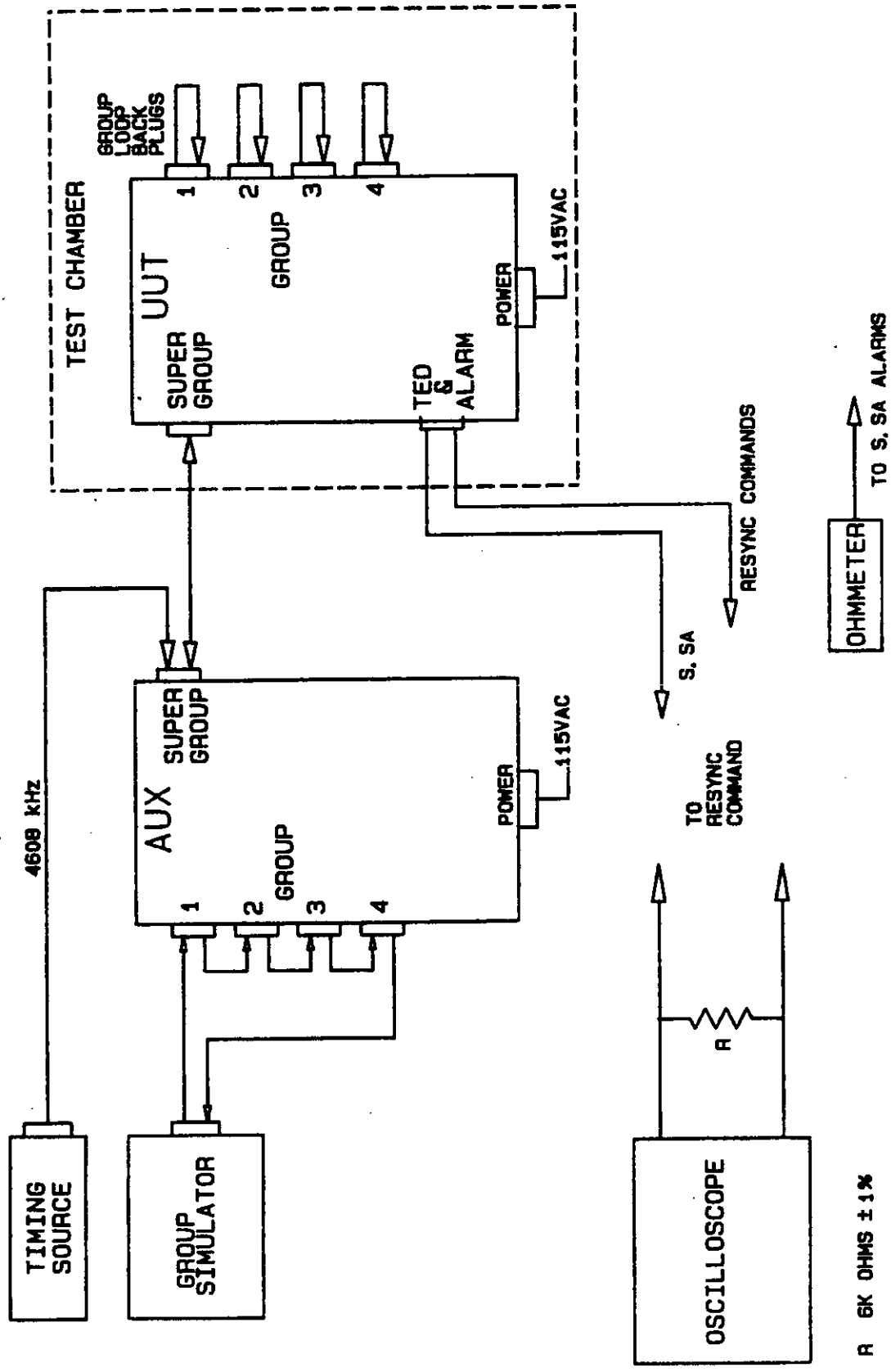


FIGURE 22. Unit performance electrical tests.

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(See Instructions – Reverse Side)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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